

Subject Name: DBMS Subject Code:3130703

**UNIT NO- 1 :
Database System Architecture**

TOPIC:1 : Introduction and applications of DBMS, Purpose of data base

Sr. No	SHORT QUESTIONS (1 Mark) / MCQ / True-False/Fill in the blanks	Marks
1	Who is a DBA? (Dec 2015)[SSIT] Answer: Database administrators (DBAs) use specialized software to store and organize data.	1
2	Define: Instance and Schema. (Jan 2017) [SSIT] Answer: Design of a database is called the schema Data stored in database at a particular moment of time is called instance of database.	1
3	Define DBMS. (Jan 2017) [SSIT] Answer: It is a collection of programs that enables user to create and maintain a database	1
4	Write advantages of Database System.(Jun 2017)[SSIT] Answer: 1. Redundancy is controlled. 2. Unauthorized access is restricted. 3. Providing multiple user interfaces. 4. Enforcing integrity constraints. 5. Providing backup and recovery	1
5	Define DBMS and list out purpose of DBMS. (Dec 2015)[SSIT] Answer: It is a collection of programs that enables user to create and maintain a database Purpose 1. Redundancy is controlled. 2. Unauthorized access is restricted. 3. Providing multiple user interfaces. 4. Enforcing integrity constraints. 5. Providing backup and recovery.	1
6	_____ is data about data. A. Data type. B. Data item. C. Meta data. D. Information. (Dec 2015)[SSIT]	1
7	Write any one difference between traditional file processing system and database management system (May 2016)[SSIT] Answer: (write any one) • Redundancy is controlled. • Unauthorized access is restricted. • Providing multiple user interfaces. • Enforcing integrity constraints. • Providing backup and recovery.	1
8	What do you mean by instances and schemas? (May 2016)[SSIT] Answer: Definition of schema: Design of a database is called the schema. Definition of instance: The data stored in database at a particular moment of time is called instance of database.	1
9	What is data dictionary? (May 2016) [SSIT]	1

	Answer: A data dictionary is a file or a set of files that contains a database's metadata. The data dictionary contains records about other objects in the database, such as data ownership, data relationships to other objects, and other data .	
10	Define : Database [SSIT]	1
11	What are the advantages of DBMS ? [SSIT]	1
12	Define : DBMS [SSIT]	1
13	_____refers to the basic facts and entities , such as names and numbers [SSIT]	1
14	Which is the false statement ? a. A Database is ordered collection of data. b. A Database is systematic compilation of records in a computer c. DBMS manages the database d. Data helps in making decisions. ANS :- d [SSIT]	1
15	What are the disadvantages in file processing system ? [SSIT]	1
16	Define Metadata.(Nov-2019_Old) [SSIT] Answer: Metadata is simply defined as data about data. It means it is a description and context of the data.	1
Sr. No	DESCRIPTIVE QUESTIONS	Marks
1	Explain different database languages (Dec2009) [SSIT]	3
2	Explain the purpose of the database system (Dec 2009) [SSIT]	4
3	Describe various disadvantages of file system compare to Data base management system. (Mar 2010) [SSIT]	7
4	Explain the purpose and application of DBMS (Dec 2011) [SSIT]	5
5	Discuss disadvantages of file processing systems and advantages of Database Management Systems (May 2012)(june 2019) [SSIT]	7
6	Explain disadvantages of conventional file-based system compared to Database management system (Dec2014, May 2014,May 2018 [SSIT]	7
7	List the benefits of database approach. (May2011) [SSIT]	2
8	Explain DBMS is better than file management system. (Dec 2015 Old) [SSIT]	5
9	Briefly explain applications of DBMS(May 2015 Old)[SSIT]	4
10	Discuss the role of Database Administrator (DBA) (May 2015 Old)[SSIT]	7
11	Briefly explain applications of DBMS (May 2016) [SSIT]	3
12	Explain characteristics/features of DBMS which differentiate it from File System Processing. (Jan 2017 Old)[SSIT]	7
13	Explain : (1) Database (2) Meta data (3) Data Dictionary (4) user_constraints table (5) check constraint (6) null (7) cursor (Jun 2017 Old)[SSIT]	7
14	Compare Following: Physical independence vs Logical independence(May 2016 Old)[SSIT]	3.5
15	What is Database Management system? Explain difference between DBMS & RDBMS.What are the possible applications of DBMS? (Dec 2015 Old)[SSIT]	7
16	Define data and information. Also explain advantages of DBMS. (May 2015 Old)[SSIT]	7
17	What is data dictionary? What it stores? How can this information be useful in DBMS. (May 2015 Old)[SSIT]	7
18	Define DBMS. Describe significant differences between a file-processing system and a DBMS.(Nov 2017) [SSIT]	4
19	Explain advantages of database management system over a file oriented system.(Dec 2018) [SSIT] Explain Advantages of Database management system over File oriented system.(Nov-	7,4

	2019_Old) [SSIT] Enlist and explain the advantages of DBMS over traditional file system.(Nov-2019) [SSIT]	
20	What do you mean by integrity constraints? Discuss various integrity constraints.(Nov-2019) [SSIT]	3
	TOPIC:2 : Data, Independence Database System architecture- levels, Mappings, Database, users and DBA	
Sr. No	SHORT QUESTIONS (1 Mark) / MCQ / True-False/Fill in the blanks	Marks
1	What is the main role of DBA? (Jan 2017) [SSIT] Answer: planning, design, development of database applications, and management of an organisation's data and metadata	1
2	What do you mean by Data Independence ? [SSIT]	1
3	Which is the type of Data Independence ? a. Physical Data Independence b. Logical Data Independence c. Both d. None of these ANS :- c [SSIT]	1
4	State Main functional Components of Database System. [SSIT]	1
5	What are the components of Storage Manager ? [SSIT]	1
6	What are the components of Query Processor ?[SSIT]	1
7	Internal level has _____ a. Individual users view of the database. b. Community view of the database. c. Physical representation of the database. d. All of these. ANS :- c [SSIT]	1
8	Schema is usually stored in _____. a. Tables b. Data Dictionary c. Both d. None of these. ANS :- b [SSIT]	1
9	_____ is data about data. A. Data type. B. Data item. C. Meta data. D. Information. ANS :- C [SSIT]	1
10	Differentiate between Database User and Database Administrator.[SSIT]	1
11	What are the functions of DBA ?[SSIT]	1
12	The Users who are easy to use menu are called _____ a. Sophisticated end user b. Naïve user c. Stand alone users d. Casual end users. ANS :- b[SSIT]	1
13	Which of the following is more concerned about the conceptual level of DBMS ? a. DBA b. The end user c. The System Programmer d. Client ANS :- a[SSIT]	1

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Sr. No	DESCRIPTIVE QUESTIONS	Marks
1	Explain functions and responsibilities of DBA. (Dec 2009) [SSIT]	4
2	What are the responsibilities of a DBA? (Dec 2010) [SSIT]	4
3	What is data independence? Explain the difference between physical and logical data independence with example. (Dec 2010) [SSIT]	5
4	List the major functions performed by DBA. (May2011) [SSIT]	2
5	Explain functions of DBA to handle DBMS. (Dec. 2011) [SSIT]	5
6	Differentiate the following: DA and DBA (May 2012) [SSIT]	4
7	Draw three level architecture of DBMS. (Dec. 2011) [SSIT]	4
8	Explain database system architecture with diagram in detail. (May2011) [SSIT]	7
9	Explain three level architecture of database system (Dec 2009, May 2012 ,Dec 2014,May 2018, Dec 2018) [SSIT]	7
10	Explain in detail the duties of DBA (May2013) [SSIT]	5
11	Define DBMS. Explain the various roles of a DBA. (Jan 2013, May 2015 Old ,Jun 2017 Old)[SSIT]	7
12	What is data independence? Explain different types of data independence with suitable example. (Nov 2013) [SSIT]	7
13	Explain database system 3 tier architecture with clear diagram in detail. (May2013, May 2015) [SSIT]	7
14	Define database. Explain the three level architecture of DBMS. (Jan 2013) [SSIT]	7
15	Explain the different types of database users. (May 2014) [SSIT]	3
16	Explain three level architecture of DBMS (May 2014)[SSIT]	7,4
17	Explain the three level architecture of DBMS and its advantages. (May 2015) [SSIT]	7
18	Explain Database Management System Architecture with proper diagram. (May 2015) [SSIT]	7
19	Discuss the role of database administrator (DBA). (May 2016 Old) [SSIT]	7
20	Explain Database Management System (DBMS) architecture with proper diagram. (May 2016 Old) [SSIT]	7
21	Define Schema. Explain three level architecture in DBMS. (Jun 2017 Old)[SSIT]	7
22	Explain database system architecture levels. (Jun 2017 Old)[SSIT]	3
23	Explain Data Independence and its types in detail. (Jun 2017 Old)[SSIT]	7
24	Explain Following: 1. Three schema architecture of DBMS. 2. Role of DBA. (Jan 2017 Old)[SSIT]	7
25	Explain tasks of DBA.(Jun 2017) (May 2018) [SSIT]	3,4
26	explain three level architecture of DBMS. (Jan 2017) [SSIT]	7,4
27	Enlist and briefly explain categories of database users. (Jan 2017) [SSIT]	4
28	Draw and discuss three level structure of database management system.(May 2016 Old)[SSIT]	7
29	List and discuss various types of database users. (May 2016 Old)[SSIT]	7
30	Explain detail duty of Data Base Administrator. (Dec 2015 Old)[SSIT]	7
31	What is the difference between Database user and Database administrator? Explain various functions of Database Administrator? (Dec 2014 Old) [SSIT]	7
32	What are main functions of a Database Administrator? (Nov 2017) [SSIT]	4
33	Explain types of database users(Dec 2018) [SSIT]	4
34	Explain Three level architecture of DBMS.(Nov-2019_Old) [SSIT]	4
35	Explain Instance and Schema in detail.(Nov-2019) [SSIT]	3
	UNIT NO- 2: Data Models	

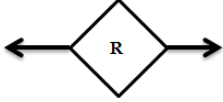
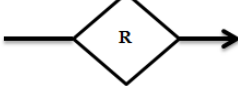
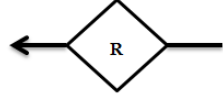
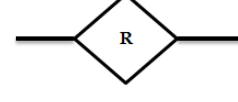
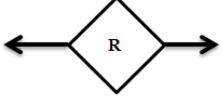
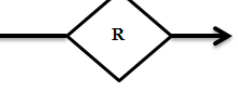
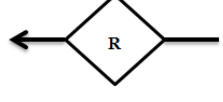
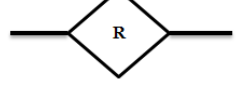
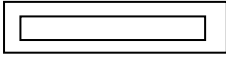
	TOPIC:1 :Structure of relational databases, Domains, Relations	
Sr. No	SHORT QUESTIONS (1 Mark) / MCQ / True-False/Fill in the blanks	Marks
1	The number of tuples in a relation is known as A. degree B. relation C. attribute D. cardinality .(Jun 2017)[SSIT] Answer: D	1
2	Which of the following aspects of data is the concern of a relational database model? A. data manipulation B. data integrity C. data structure D. all of these . (Jun 2017)[SSIT] Answer: B	1
3	A relational database consists of a collection of _____. a) Tables b) Fields c) Records d) Keys (Jan 2017) [SSIT] Answer:a	1
4	Define RDBMS (May 2016)[SSIT] Answer: A relational database management system (RDBMS) is a database management system (DBMS) that is based on the relational model as invented by E. F. Codd.	1
5	What is a candidate key? Define candidate key. (May 2016)[SSIT] Answer: A candidate key is a column, or set of columns, in a table that can uniquely identify any database record without referring to any other data.	1
6	Define NULL constraint. (May 2016)[SSIT] Answer: The NULL is the term used to represent a missing value	1
7	What is the relational model ? [SSIT]	1
8	Define term Relation.[SSIT]	1
9	Define : Domain.[SSIT]	1
10	Define Cardinality and degree of the relation.[SSIT]	1
11	What is referential integrity ?[SSIT]	1
12	A Set of Possible data value is called _____. a. Attribute b. Degree c. Tuple d. Domain ANS :- d [SSIT]	1
13	In the relational modes , Cardinality is termed as _____. a. Number of Tuples b. Number of Attributes c. Number of tables d. Number of Constraints. ANS :- a [SSIT]	1
Sr. No	DESCRIPTIVE QUESTIONS	Marks
1	Explain candidate key, primary key and foreign key (Dec 2009) [SSIT]	3
2	Define: (1) Primary key (2) Foreign key (3) Unique Key (4) Not null (5) Commit (6) Candidate key (7) Rollback. (May 2013) [SSIT]	7
3	Explain following terms with suitable example. (1) Primary Key (2) Candidate Key (3) Foreign Key (4) On Delete Cascade (Dec 2013) [SSIT]	7
4	Define: (1) Primary key (2) Foreign key (3) Unique Key (4) Not null (5) Commit (6)	7

	Candidate key (7) Rollback (Jan 2013) [SSIT]	
5	Compare following: 1. Primary key vs Foreign key(May 2016 Old)[SSIT]	3.5
6	Explain candidate key, primary key and foreign key with example. (Dec 2015 Old)[SSIT]	7
7	List and explain various data models used for database design. (May 2015 Old)[SSIT]	7
8	Define Primary key, Candidate key and Super key. (Nov 2017)(Nov-2019) [SSIT]	3
9	List reasons why null values might be introduced into the database. (Nov 2017) [SSIT]	3
	TOPIC:2 : Relational algebra – fundamental operators and syntax, relational algebra queries, tuple relational calculus	
Sr. No	SHORT QUESTIONS (1 Mark) / MCQ / True-False/Fill in the blanks	Marks
1	Write a symbol for project operation in relational algebra. (May 2016)[SSIT] Answer: Project Operation (⌋) It projects column(s) that satisfy a given predicate.	1
2	What is relational algebra ? [SSIT]	1
3	What are the unary operations in relational algebra ?[SSIT]	1
4	What is degree of a relation ?[SSIT]	1
5	Set Concept is used in _____ a. Network Model b. Hierarchical Model c. Relational Model d. None of these. ANS :- c [SSIT]	1
6	Relational Algebra is _____. a. Data definition language b. Meta language c. Procedural Query language d. None of the above. ANS :- c [SSIT]	1
7	What is Relational Calculus ? [SSIT]	1
8	How does Tuple oriented relational Calculus differ from domain oriented relational calculus ? [SSIT]	1
9	What are different types of relational Constraints ?[SSIT]	1
Sr. No	DESCRIPTIVE QUESTIONS	Marks
1	Explain following relational algebra operation (i)Natural join operation (ii) Selection and projection operation (Dec 2009) [SSIT]	4
2	What is join? Explain various type of joins with example (Dec 2010) [SSIT]	7
3	Consider following schema and represent given statements in relation algebra form. * Branch(branch_name,branch_city) * Account(branch_name, acc_no, balance) *Depositor(Customer_name, acc_no) (i) Find out list of customer who have account at 'abc' branch. (ii) Find out all customer who have account in 'Ahmedabad' city and balance is greater than 10,000. (iii) find out list of all branch name with their maximum balance. (Mar 2010) [SSIT]	6
4	List relational algebra operators and explain any two with example (May 2011) [SSIT] List the relational algebra operators. Discuss any two such algebra operator with suitable	4, 5

	example. (Nov-2019) [SSIT]	
5	Explain selection and projection operation with example. (Dec 2011) [SSIT]	5
6	Explain natural join operation with example. (Dec 2011) [SSIT]	5
7	List relational algebra operators and explain any two with example (May 2011) [SSIT]	5
8	What is join? Explain various type of joins with example (May 2012, Jun 2017 Old) [SSIT]	4 / 7
9	What is Relational Algebra? Define Relational Algebra Operation cross product with example. (May 2012) [SSIT]	3
10	What is NULL? Explain (May 2012) [SSIT]	2
11	What is database schema? Explain the select, project, natural join, union and Cartesian product operations. (Jan 2013) [SSIT]	7
12	Explain various types of join with suitable example (May2013, Nov2013) [SSIT]	7
13	List all the Relational algebra operators. Explain the working of Cartesian product Operation and the Division Operation with an appropriate example. (May 2014, Dec 2014) [SSIT]	3
14	Write short note on Relational algebra. (May 2014) [SSIT]	7
15	What is a Join? Explain different types of outer join with appropriate example (May 14)[SSIT]	5
16	Explain different types of outer join with example (May 2014, May 2018, Dec 2018) [SSIT]	7
17	Define: (1) Data (2) Entity (3) Meta Data (4) Super key (5) Not Null (6) Data Integrity (7) Dual (May 2015) [SSIT]	7
18	Solve the queries for the following database using Relational Algebra branch (branch-name, branch-city, assets) customer (customer-name, customer-street, customer-only) account (account-number, branch-name, balance) loan (loan-number, branch-name, amount) depositor (customer-name, account-number) borrower (customer-name, loan-number) 1) Find all loans of over \$1200 2) Find the loan number for each loan of an amount greater than \$1200 3) Find the names of all customers who have a loan, an account, or both, from the bank 4) Find the names of all customers who have a loan and an account at bank. 5) Find the names of all customers who have a loan at the Perryridge branch. 6) Find the names of all customers who have a loan at the Do not have an account at any branch of the bank. 7) Find the names of all customers who have a loan & an account at the Perryridge Branch. (Dec 2015) [SSIT]	7
19	Explain all types of Joins with commands and examples. (May 2015) [SSIT]	7
20	List the relational algebra operators. Discuss any one such algebra operator with suitable example. (May 2016) [SSIT]	3
21	Discuss following terms with proper example(s). 1. Referential Integrity (Foreign Key) constraint 2. Check constraint (May 2016) [SSIT]	4
22	Consider the following relational database, where the primary keys are underlined. Give an expression in the relational algebra to express each of the following queries: employee (ssn, name, dno, salary, hobby, gender) department (dno, dname, budget, location, mgrssn) works_on (ssn, pno) project (pno, pname, budget, location, goal) 1. List all pairs of employee names and the project numbers they work on. 2. List out department number, department name and department budget. 3. List all projects that Raj Yadav works on by project name.	4

	4. List the names of employees who supervise themselves.(Jun 2017) [SSIT]	
23	Explain Cartesian Product operation in Relational ALGEBRA.(Jan 2017 Old)[SSIT]	4
24	Explain Foreign Key constraint in Relational Model with example.(Jan 2017 Old)[SSIT]	4
25	Explain various types of JOIN with example. (Jan 2017 Old)[SSIT]	8
26	What is join? List and explain various types of joins. (Jan 2017) [SSIT]	3
27	Discuss various types of joins with suitable example. (May 2016 Old)[SSIT]	7
28	(b) Consider following schema and represent given statements in relation algebra form. • Branch(branch_name,branch_city) , • Account(branch_name, acc_no,balance) • Depositor(Customer_name, acc_no) 1. Find out list of customer who have account at 'Baroda HO' branch. 2. Find out all customers who have account in 'Ahmedabad' city and balance is greater than 10,000. 3. Find out list of all branch names with their maximum balance. (Dec 2015 Old)[SSIT]	7
29	What is join? Explain various types of joins with example. (Dec 2015 Old)[SSIT]	7
30	Write SQL or Relational algebra statement for following. Student(Enrno, name, courseId, emailId, cellno) Course(courseId, course_nm, duration) 1) Add a column city in student table. (1) 2) Find out list of students who have enrolled in "computer" course. (2) 3) List name of all courses with their duration. (1) 4) List name of all students start with 'a'. (1) 5) List emailId and cellno of all mechanical engineering students. (2) (May 2015 old)[SSIT]	7
31	What is relational algebra? List and explain Fundamental and additional relational algebra operators. (May 2015 Old)[SSIT]	7
32	What is Relational Algebra? Enlist and Explain the fundamental operation of relational algebra. (Dec 2014 Old) [SSIT]	7
33	What is Relational Algebra? Define Relational Algebra Operation Cross product with example. (Nov 2017) [SSIT]	3
34	Consider the relational database given below. Give an expression in the relational algebra to express each of the following queries: Employee (person-name, street, city) , Works (person-name, company-name, salary) Company (company-name, city) , Manages (person-name, manager-name) 1) Find the names of all employees in this database who live in the same city as the company for which they work. 2) Find the names, street address, and cities of residence of all employees who work for HCL and earn more than \$10,000 per annum. (Nov 2017) [SSIT]	4
35	Explain Outer Joins operations. (Nov 2017) [SSIT]	3
36	Explain following relational algebraic operation (i) Division (ii) inner join (iii) intersection (june 2019)[SSIT]	07
37	Explain following relation algebra operations: (i) full outer join (ii) projection (iii) Cartesian product. (June 2019)[SSIT]	07
38	The relational database schema is given below. employee (person-name, street, city) works (person-name, company-name, salary) company (company-name, city) manages (person-name, manager-name) Write the relational algebra expressions for the given queries. 1.Find the names of all employees who work for First Bank Corporation. 2.Find the names and cities of residence of all employees who work for First Bank Corporation. 3.. Find the names, street address, and cities of residence of all employees who work for First Bank Corporation and earn more than \$10,000 per annum. 4. Find the names of all employees in this database who do not work for First Bank	04

	Corporation.(Nov-2019)[SSIT]	
	UNIT NO- 3: Relational query languages:	
	TOPIC:1 : Basic concepts, Design process, constraints, Keys, Design issues	
	SHORT QUESTIONS	
1	What is E-R Model ?[SSIT]	1
2	Differentiate between Condition defined and user defined Constraints.[SSIT]	1
3	Differentiate between Disjoint and Overlapping Constraints.[SSIT]	1
4	What do you mean by Key of the relation ?[SSIT]	1
5	Differentiate between Super Key and Candidate Key.[SSIT]	1
Sr. No	DESCRIPTIVE QUESTIONS	Marks
1	With example explain various mapping cardinalities and total participation(Mar 2010)[SSIT] OR Explain various mapping cardinalities.(Nov 2019) [SSIT]	7 3
2	Define: (1) Data (2) Entity (3) Meta Data (4) Super key (5) Not Null (6) Data Integrity (7) Dual (May 2013) [SSIT]	7
3	Explain Mapping Cardinalities. (May 2012) [SSIT]	2
4	Explain following terms with suitable example (1) Primary Key (2) Candidate Key (3) Foreign Key (4) On Delete Cascade (Nov 2013) [SSIT]	7
5	Define: (1) Weak entity (2) Data Dictionary (3) Substring() (4) Dual (5)Alter (6) Truncate (7) Drop. (Jan 2013) [SSIT]	7
6	What is constraint in database? Explain types of constraints with suitable example May 2013 (Dec 2014) [SSIT]	5
7	Draw an E-R diagram for Banking enterprise. 07 (Dec 2014 Old) [SSIT]	7
8	What are integrity constraints? Explain various types of integrity constraints with suitable example (May 2014) [SSIT]	7
9	Explain mapping cardinalities and total participation with appropriate example. (Dec 2015 Old)[SSIT]	7
10	Write a note on mapping cardinality in E-R Diagram (May 2016 Old) [SSIT]	4
11	Explain following terms with reference to E-R model. 1. Entity & it's Attributes. 2. Relationship Type (including recursive). 3. Cardinality & Participation. 4. Weak-Entity Type & Relationship Type (Jan 2017 Old)[SSIT]	8
	TOPIC:2 : E-R diagrams, weak entity sets, extended E-R features – generalization, specialization, aggregation,	
Sr. No	SHORT QUESTIONS (1 Mark) / MCQ / True-False/Fill in the blanks	Marks
1	_____ are made of more than one simple attribute. (Jan 2017) [SSIT] Answer: Composite attribute	1
2	Define entity set and attribute in E-R diagram. (May 2016)[SSIT] Answer: Entity set is a collection of all entities of particular entity type in the database. Entities are represented by means of their properties, called attributes . All attributes have values.	1

3	List various mapping cardinalities of E-R diagram. (Jan 2017) [SSIT] Answer:  ONE-TO-ONE CARDINALITY  MANY-TO-ONE CARDINALITY  ONE-TO-MANY CARDINALITY  MANY-TO-MANY CARDINALITY	1
4	List out the Mapping Cardinalities in ER diagram (Dec 2015)[SSIT] Answer:  ONE-TO-ONE CARDINALITY  MANY-TO-ONE CARDINALITY  ONE-TO-MANY CARDINALITY  MANY-TO-MANY CARDINALITY	1
5	Draw symbols for following in E-R diagram: Weak Entity set, Derived attribute. (Dec 2015)[SSIT] Answer: Weak Entity set  Derived attribute 	1
6	Give Various types of Relationships.[SSIT]	1
7	Differentiate between Unary and Binary Relationship.[SSIT]	1
8	Define Weak and Strong Entity Sets.[SSIT]	1
9	What does the Cardinality ratio specify?[SSIT]	1
10	What is an EER?[SSIT]	1
11	What is Specialization?[SSIT]	1
12	What is Generalization?[SSIT]	1
13	What is Aggregation?[SSIT]	1
Sr. No	DESCRIPTIVE QUESTIONS	Marks
1	Explain specialization and generalization feature of ER diagram with example (Dec 2009,)[SSIT]	3
2	Explain aggregation operation of ER diagram. (Dec 2009) [SSIT]	3
3	Construct E-R diagram for a hospital with a set of patients and medical doctors. associate with each patient a log of various tests and examinations conducted (Dec 2009) [SSIT]	4
4	Construct E-R diagram of the bank. It provides different kinds of bank accounts. And loans. It operates number of branches. (Dec 2009) [SSIT]	4
5	Explain specialization and generalization feature of ER diagram with example (Dec 2009) [SSIT]	3
	Explain specialization and generalization concepts in ER diagram with suitable example. May 2018, Dec 2018 [SSIT]	07

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6	Explain aggregation operation of ER diagram. (Dec 2009) [SSIT]	3
7	Construct E-R diagram for a hospital with a set of patients and medical doctors. associate with each patient a log of various tests and examinations conducted (Dec 2009) [SSIT]	4
	Construct E-R diagram of the bank. It provides different kinds of bank accounts. And loans. It operates number of branches. (Dec 2009) [SSIT]	4
8	Draw E-R diagram for Hospital management system and covert into set of table schema (Mar 2010) [SSIT]	7
9	Draw E – R Diagram for the School Management System. (Dec 2010) [SSIT]	7
10	Draw E – R Diagram for Library Management System. (Dec 2010) [SSIT]	7
11	With example explain various mapping cardinalities and total participation (Mar 2010) [SSIT]	7
12	Draw E-R diagram for Hospital management system and covert into set of table schema (Mar 2010) [SSIT]	7
13	Give Symbol used in E-R Diagram and Draw the E-R diagram of Library Management System. (Dec 2011) [SSIT]	7
14	Give Symbol used in E-R Diagram and Draw the E-R diagram of University exam System (Dec 2011) [SSIT]	7
15	Draw E-R diagram for supplier who supplies different parts. The parts are used in different projects. Explain the mapping cardinality used. Assume suitable attributes. (May 2011) [SSIT]	5
16	Explain aggregation operation of E-R Diagram. (Dec 2011) [SSIT]	5
17	Explain generalization feature of E-R Diagram. (Dec 2011) [SSIT]	5
18	What does ER model mean? Specify all its notations. (May 2012) [SSIT]	2
19	Explain generalization and specialization in ER diagram with suitable example . (May 2011, May2013) [SSIT]	4
20	Construct an E-R Diagram for an insurance company with a set of customers, each of whom owns number of cars, also each can have number of recorded accident associated with it. (May 2012) [SSIT]	5
21	Obtain E-R diagram for the Admission procedure in a university. An advertisement is issued giving essential qualifications for the course, the last date for recipient of application, and the fee to be enclosed with the application. A clerk in the registrar's office checks the received applications to see if mark sheet and the fee are enclosed and sends valid application to the concerned academic department. The department checks the application in detail and decides the applicant to be admitted, those to be put in the waiting list, and those to be rejected. Appropriate letters are sent to the registrar's office which intimates the applicant. (May 2012) [SSIT]	5
22	Draw symbols for following in ER diagram Explain Relationship Set, and Primary key attribute (May 2013) [SSIT]	2
23	Draw ER diagram for Library Management System. (May 2013) [SSIT]	7
24	Construct E-R diagram for a hospital with a set of patients and medical doctors. Associate with each patient a log of various tests suggested by doctors and examinations conducted. Use Specialization and Generalization in your diagram. (Nov 2013) [SSIT]	7
25	Construct E-R diagram for a bank which has many branches and it supports different types of accounts. It also provides loans to customers. Use Specialization and Generalization in your diagram. (Nov 2013) [SSIT]	7
26	Explain Entity Relationship model with example. (May 2014) [SSIT]	7
27	Draw an Entity Relation diagram for the Hospital Management System. Consider the different types of Patients with respect to Disease and In-Patient and Out-Patient Department in the design. Consider the availability of all well qualified Doctors. Consider various types of tests and operations to be conducted. Explain the mapping cardinality used. Assume suitable attributes. Use generalization and Specialization. (Dec 2014) [SSIT]	7

Shree Swaminarayan Institute of Technology Semester: III (2020)

28	Explain Entity Relationship model with example. (May 2014) [SSIT]	7
29	Draw an ER diagram for describing the activities of a departmental store (Dec 2015) [SSIT]	7
30	Define E-R Diagram. Draw E-R diagram with Customer, Loan and Payment sets. (May 2015) [SSIT]	7
31	Draw an ER diagram for a car insurance company that has a set of customers each of whom owns one or more cars. Each car has associated with it 0 to any number of recorded accidents. (Dec 2015) [SSIT]	7
32	Signify the concept of Aggregation in ER Diagram with example (May 2015 Old) [SSIT]	3
33	Define E-R diagram. Draw an E-R diagram for Library Management System. Assume relevant entities and attributes for the given system. (May 2016) [SSIT]	7
34	Explain Specialization, Generalization and Categorization in DBMS. (Jun 2017 Old) [SSIT]	7
35	(I) In college database, we have following: • A college contains many departments • Each department can offer any number of courses • Many instructors can work in a department • An instructor can work only in one department • For each department there is a Head • An instructor can be head of only one department • Each instructor can take any number of courses • A course can be taken by only one instructor • A student can enroll for any number of courses • Each course can have any number of students Draw an ER diagram for the data set described above. Make sure to indicate all cardinality constraints specified above. The ER diagram should not contain redundant entity sets, relationships, or attributes. Also, use relationships whenever appropriate. (II) A weak entity set can always be made into a strong entity set by adding to its attributes the primary-key attributes of its identifying entity set. Outline what sort of redundancy will result if we do so.(Jun 2017) [SSIT]	7
36	In academic world, A researcher can either be employed as a professor or a lab assistant. There are three kinds of professors: Assistant, associate, and full professors. The following should be stored: • For each researcher, his/her name, year of birth, and current position (if any). • For each institution, its name, country, and inauguration year. • For each institution, the names of its schools (e.g. School of Law, School of Business, School of Computer Science, . . .). A school belongs to exactly one institution. • An employment history, including information on all employments (start and end date, position, and what school). • Information about co-authorships, i.e., which researchers have co-authored a research paper. The titles of common research papers should also be stored. • For each researcher, information on his/her highest degree (BSc, MSc or PhD), including who was the main supervisor, and at what school. • For each professor, information on what research projects (title, start date, and end date) he/she is involved in, and the total amount of grant money for which he/she was the main applicant. Draw an ER diagram for the data set described above. Make sure to indicate all cardinality constraints specified above. The ER diagram should not contain redundant entity sets, relationships, or attributes. Also, use relationships whenever appropriate.(Jun 2017) [SSIT]	7
37	Define E-R diagram. Discuss generalization and specialization in E-R diagram with suitable diagram. (Jan 2017) [SSIT]	7
38	Draw E-R diagram for bank management system. (Jan 2017 , May 2018) [SSIT]	7
39	Define E-R Diagram. List and explain various components of E-R diagram. Draw and	7

	explain generalization and specialization E-R diagram for Banking Management System. Assume relevant entities and attributes.(May 2016 Old)[SSIT]	
40	Write functionalities that are required for making one School Management System in brief. Draw ER Diagram of School Management System. (Dec 2015 Old)[SSIT]	7
41	Draw an ER diagram for Event Management System. (May 2015 Old)[SSIT]	7
42	What is Entity set? Also define Relationship set. List and explain all the symbols used to draw ER Diagram. (May 2015 Old)[SSIT]	7
43	Explain the difference between a weak and a strong entity set. (Nov 2017) [SSIT]	3
44	Design a generalization–specialization hierarchy for a motor-vehicle sales company. The company sells motorcycles, passenger cars, vans, and buses. Justify your placement of attributes at each level of the hierarchy. Explain why they should not be placed at a higher or lower level. (Nov 2017) [SSIT]	7
45	Construct an E-R diagram for a car-insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents. (Nov 2017) [SSIT]	7
46	Explain difference between strong entity set and weak entity set.(Dec 2018) [SSIT]	04
47	Define E-R diagram. Draw an E-R diagram for Library Management System. Assume relevant entities and attributes for the given system.(Dec 2018) [SSIT]	07
48	Design a database for an airline. The database must keep track of customers and their reservations, flights and their status, seat assignments on individual flights, and the schedule and routing of future flights. Your design should include an E-R diagram, a set of relational schemas, and a list of constraints, including primary-key and foreign-key constraints (june 2019)[SSIT]	07
49	Design a database for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted Your design should include an E-R diagram, a set of relational schemas, and a list of constraints, including primary-key and foreign-key constraints. (june 2019)[SSIT]	07
50	Define E-R Diagram. Draw an E-R Diagram for Hospital Billing System.(Nov-2019_Old) [SSIT]	7
51	What are the different types of relationships in the DBMS?(Nov-2019_Old) [SSIT]	3
52	Construct an E-R diagram for a car insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents. Each insurance policy covers one or more cars, and has one or more premium payments associated with it. Each payment is for a particular period of time and has an associated due date and the date when the payment was received.(Nov-2019) [SSIT]	7
53	Explain specialization and generalization concepts in ER diagram with suitable example.(Nov-2019) [SSIT]	7
	TOPIC:3 : Reduction to E-R database schema	
Sr. No	DESCRIPTIVE QUESTIONS	Marks
1	What is Entity-Relationship model? Explain the steps to reduce the ER diagram to ER database schema. (Jan 2013) [SSIT]	7
2	List the steps in proper sequence in order to convert an ER and EER diagram into tables. (May 2015) [SSIT]	7
	UNIT NO – 4: Relational Database design	
	TOPIC:1 Functional Dependency – definition, trivial and non-trivial FD, closure of FD set, closure of attributes, irreducible set of FD	

Sr. No	SHORT QUESTIONS (1 Mark) / MCQ / True-False/Fill in the blanks	Marks
1	Give an example of a relation schema R and a set of dependencies such that R is in BCNF but is not in 4NF. (Jun 2017) [SSIT] Answer: R(A, B, C) A →→ B	1
2	Considering functional dependency, one in which removal from some attributes must affect dependency is called A. full functional dependency B. partial dependency B. prime functional dependency D. transitive dependency (Jun 2017) [SSIT] Answer: A	1
3	List the Pitfalls of Relational Database System. [SSIT]	1
4	Define : Functional Dependency .[SSIT]	1
5	What are the uses of Functional Dependency ? [SSIT]	1
6	What is Fully Functional dependency ? [SSIT]	1
7	What is closure of a Set of Functional Dependencies ? [SSIT]	1
8	What are Axioms ? [SSIT]	1
9	What is Canonical cover ? [SSIT]	1
Sr.no.	DESCRIPTIVE QUESTIONS	Marks
1	What are anomalies in database design? How can we solve it? (Dec 2009 Dec 2013) [SSIT]	3
2	Explain how to find closure of a set of attributes? (Dec 2009) [SSIT]	3
3	What is non-loss decomposition in database? How it is useful in database? (Dec 2009 Dec 2013) [SSIT]	3,4
4	What is canonical cover? Consider following set F of functional dependencies on schema R(A,B,C) and compute canonical cover for F. A → BC B → C A → B AB → C. (Mar 2010)(June 2019)[SSIT] [SSIT]	6,5
5	Consider a relation R with five attribute A,B,C,D,E having following dependencies : A → B, BC → E and ED → A a) List all Keys for R b) In which normal form table is, justify your answer. Dec 2010 [SSIT]	7
6	State and explain Heath's Theorem. May 2011 [SSIT]	2
7	Given relation R with attributes A,B, C,D,E,F and set of FDs as A → BC, E → CF, B → E and CD → EF. Find out closure {A, B} + of the set of attributes. May 2011 [SSIT]	3
8	Define functional dependency. Explain trivial and non-trivial FD with example. May 2011 [SSIT]	3
9	Explain irreducible sets of Functional dependency with example. May 2012 [SSIT]	3
10	What problems can occur due to wrong database design? How they can be solved? May 2011 (Dec 2014) [SSIT]	4
11	What is Functional Dependency? Explain non-loss decomposition. (May 2012) [SSIT]	7
12	Consider table R(A,B,C,D,E) with FDs as A → B, BC → E and ED → A. The table is in which normal form? Justify your answer (May 2013) [SSIT]	7
13	Explain the issues of a database design. (Jan 2013) [SSIT]	7
14	What is functional dependency? Explain its usage in database design. (May 2013) [SSIT]	3
15	What is Functional Dependency? Explain. (Dec 2014) [SSIT]	2
16	What is decomposition? Why is it required? Explain the difference between lossy decomposition and non-loss decomposition with example (May 2014) [SSIT]	5
17	What is functional dependency? Explain its types in detail. (May 2015) [SSIT] What is Functional dependency? Explain all its types with example. (Nov-2019_Old) [SSIT]	7 7
18	A college keeps details about a student and the various modules the student studied. These details comprise regno - registration number , n - student name , a - student address , tno - tutor number , tna - tutor name , dc - diploma code , dn - diploma name , mc - module code , mn - module name , res - module exam result where details(regno, n,a, tno, tna,dc ,dn ,(mc	4

	,mn, res)) dc -> dntno -> tna , mc, mn -> res , n -> a, mc -> mn Reduce the relation to third normal form (Dec 2015) [SSIT]	
19	Given R= (A, B, C, G, H, I). The following set F of functional dependencies holds $A \rightarrow B$ $A \rightarrow C$ $CG \rightarrow H$ $CG \rightarrow I$ $B \rightarrow H$ Compute AG^+ . Is AG a candidate key? (Dec 2015) [SSIT]	3
20	Compute the closure of R (A, B, C, D, E) with the following set of functional dependencies $A \rightarrow BC$ $CD \rightarrow E$ $B \rightarrow D$ $E \rightarrow A$ List the candidate keys of R. (Dec 2015) [SSIT]	3
21	Given relation R with attributes A,B, C,D,E,F and set of FDs as $A \rightarrow BC$, $E \rightarrow CF$, $B \rightarrow E$ and $CD \rightarrow EF$. Find out closure $\{A,B\}^+$ of the set of attributes (Dec 2015) [SSIT]	3
22	What is decomposition? Explain Lossy and Non-loss decomposition with suitable example. (Dec 2014 Old) [SSIT]	7
23	What is functional dependency and explain its various types? (Jun 2017 Old) [SSIT]	7
24	What is Functional Dependency? How is it eliminated in Data? (Jan 2017 Old) [SSIT]	6
25	Compute the closure of the following set F of functional dependencies for relation schema r (A, B, C, D, E). $A \rightarrow BC$ $CD \rightarrow E$ $B \rightarrow D$ $E \rightarrow A$ List the candidate keys for R. ? (Jun 2017) [SSIT]	7
26	Consider the following relational database, where the primary keys are underlined. Give an expression in the relational algebra to express each of the following queries: course (course-id, title, dept_name, credits) instructor (id, name, dept_name, salary) section (course-id, sec-id, semester, year, building, room_no, time_slot_id) teaches (id, course-id, sec-id, semester, year) 1. Find the name of all instructors in the physics department. 2. Find all the courses taught in the fall 2009 semester but not in Springsemester. 3. Find the names of all instructors in the Comp. Sci. department togetherwith the course titles of all the courses that the instructors teach. 4. Find the average salary in each department. (Jun 2017) [SSIT]	4
27	What is Redundancy? Explain the anomalies in relational database. (Jun 2017 Old) [SSIT]	7
28	Define decomposition. Give the differences between lossy and lossless decomposition with suitable example. (May 2016 Old) [SSIT]	7
29	What do you understand by functional dependency? Write a simple example to explain functional dependency. (Dec 2015 Old) [SSIT]	7
30	Define followings: Functional dependency, Trivial functional dependency, multivalve functional dependency, 1NF, BCNF, 3NF, 4NF. (May 2015 Old) [SSIT]	7
31	Consider schema R = (A, B, C, G, H, I) and the set F of functional dependencies $\{A \rightarrow B, A \rightarrow C, CG \rightarrow H, CG \rightarrow I, B \rightarrow H\}$. (Use F^+) Prove that $AG \rightarrow I$ Holds. (Nov 2017) [SSIT]	3
32	Consider schema EMPLOYEE(E-ID,E-NAME,E-CITY,E-STATE) and FD = {E-ID->E-NAME, E-ID->E-CITY, E-ID->E-STATE, E-CITY->E-STATE} 1) Find attribute closure for: (E-ID)+ 2) Find(E-Name)+ (Nov 2017) [SSIT]	4
33	What is redundant functional dependency? Explain trivial and non trivial functional dependency with example (june 2019) [SSIT]	07
34	Compute the closure of the following set F of functional dependencies for relation schema R = (ABCDE), F = {A -> C, E -> D, B -> C} List the candidate keys for R. (Nov-2019 Old) [SSIT]	7

35	Consider schema R = (A, B, C, G, H, I) and the set F of functional dependencies {A → B, A → C, CG → H, CG → I, B → H}. Prove that AG → I Holds.(Nov 2019) [SSIT]	4															
	TOPIC:2 : Normalization – 1Nf, 2NF, 3NF, Decomposition using FD-dependency preservation, BCNF																
	SHORT QUESTIONS	1															
1	What is Normalization ?) [SSIT]	1															
2	What is the purpose of normalization ?) [SSIT]	1															
3	List the desirable properties of Decomposition.) [SSIT]	1															
4	What do you mean by Lossless Decomposition.) [SSIT]	1															
Sr. No	DESCRIPTIVE QUESTIONS	Marks															
1	Explain 3NF with example(Dec 2009) [SSIT]	3															
2	What is normalization? What is the need for normalization?(Dec 2009 ,Dec 2014) [SSIT]	4															
3	Explain BCNF with example (Dec 2009, Dec 2013) [SSIT]	4															
4	Explain 1NF, 2NF, 3NF and BCNF (Dec 2010) [SSIT]	7															
5	Normalize (decompose) following relation into lower to higher normal form. (From 1NF to 4 NF) (Mar 2010) [SSIT] <table border="1"><tr><td>PLANT</td><td>MANAGER</td><td>MACHINE</td><td>SUPPLIER NAME</td><td>SUPPLIER CITY</td></tr><tr><td>Plant-A</td><td>Ravi</td><td>Lath Boiler</td><td>Jay industry Abb appliance</td><td>Ahmedabad Surat</td></tr><tr><td>Plant-B</td><td>Meena</td><td>Cutter Boiler CNC</td><td>Raj machinery Daksh industry Jay industry</td><td>Vadodara Rajkot Ahmedabad</td></tr></table>	PLANT	MANAGER	MACHINE	SUPPLIER NAME	SUPPLIER CITY	Plant-A	Ravi	Lath Boiler	Jay industry Abb appliance	Ahmedabad Surat	Plant-B	Meena	Cutter Boiler CNC	Raj machinery Daksh industry Jay industry	Vadodara Rajkot Ahmedabad	8
PLANT	MANAGER	MACHINE	SUPPLIER NAME	SUPPLIER CITY													
Plant-A	Ravi	Lath Boiler	Jay industry Abb appliance	Ahmedabad Surat													
Plant-B	Meena	Cutter Boiler CNC	Raj machinery Daksh industry Jay industry	Vadodara Rajkot Ahmedabad													
6	What is normalization? What is redundancy? Compare 1NF and 2NF with example. (Dec 2011) [SSIT]	7															
7	What is the advantage of using 3NF? Explain with example(Dec 2011) [SSIT]	5															
8	Explain BCNF with example (Dec 2011 ,May 2011, May 2013) [SSIT]	4															
9	Explain 3NF with example. (May 2011) [SSIT]	3															
10	What is normalization? Explain 1NF, 2NF & 3NF. (May 2012) [SSIT]	7															
11	What is normalization? Normalize following relation up to 3NF. Bank(acno, cust_name, ac_type, bal, int_rate, cust_city, branchId, branch_nm, br_city) (May 2015 Old) [SSIT]	7															
12	Explain with suitable example, the process of normalization covering from 1NF to 3NF. (Nov 2013) [SSIT]	7															
13	Why should normalization be performed on a table and what are its benefits. Explain 3NF and BCNF. (Jan 2013)(june 2019) [SSIT]	7															
14	Explain 2NF with example (May 2013) [SSIT]	3															
15	What is normalization? Why normalization process is needed? Explain 1 NF, 2 NF and 3 NF with example. (May 2014) [SSIT]	7															
16	What is Normalization? What are its characteristics? Explain 2NF and 3NF in detail. (May 2015) [SSIT]	7															
17	Prove the statement “Every relation which is in BCNF is in 3NF but the converse is not true” (Dec 2015) [SSIT]	4															
18	Elaborate Normalization. Explain any three normal forms with suitable example(s). (May 2016) [SSIT] Elaborate Normalization. Explain any two normal forms with suitable example(s). (May 2018) [SSIT]	7 04															
19	Explain 3NF & BCNF. What is the difference between them? (Jan 2017 Old) [SSIT]	8															

20	A college maintains details of its lecturers' subject area skills. These details comprise: Lecturer Number, Lecturer Name , Lecturer Grade , Department Code , Department Name , Subject Code , Subject Name , Subject Level Assume that each lecturer may teach many subjects but may not belong to more than one department. Subject Code, Subject Name and Subject Level are repeating fields. Normalise this data to Third Normal Form.(Jun 2017)(Nov 2019) [SSIT]	4, 7																								
21	What is Normalization and its properties? Explain 3NF and BCNF.(Jun 2017 Old) [SSIT]	7																								
22	In the BCNF decomposition algorithm, suppose you use a functional dependency $\alpha \twoheadrightarrow \beta$ to decompose a relation schema $r(\alpha, \beta, \gamma)$ into $r_1(\alpha, \beta)$ and $r_2(\alpha, \gamma)$. 1. What primary and foreign-key constraint do you expect to hold on the decomposed relations? 2. Give an example of an inconsistency that can arise due to an erroneous update, if the foreign-key constraint were not enforced on the decomposed relations above. 3. When a relation is decomposed into 3NF, what primary and foreign key dependencies would you expect will hold on the decomposed..(Jun 2017 Old) [SSIT]	7																								
23	A software contract and consultancy firm maintains details of all the various projects in which its employees are currently involved. These details comprise: Employee Number , Employee Name , Date of Birth , Department Code , Department Name , Project Code , Project Description , Project Supervisor Assume the following: • Each employee number is unique. • Each department has a single department code. • Each project has a single code and supervisor. • Each employee may work on one or more projects. • Employee names need not necessarily be unique. • Project Code, Project Description and Project Supervisor are repeating fields. Normalise this data to Third Normal Form.(Jun 2017) [SSIT]	4																								
24	What is meant by normalization? Write its need. List and discuss various normalization forms. (Jan 2017) [SSIT]	7																								
25	Define Normalization. List and explain various rules (forms) of normalization with proper example.(May 2016 Old) [SSIT]	7																								
26	Explain Normalization. Apply all the Normalize form to following studentsdetail table & write the resulting table. <table border="1"><tr><td>RollNo</td><td>Student_ Name</td><td>Address</td><td>Phone_ Number</td><td>Marks_ Maths</td><td>Marks_ Physics</td><td>Books_ Taken*</td></tr></table> * Books_ Taken: Books Taken by Students from Library (Dec 2015 Old) [SSIT]	RollNo	Student_ Name	Address	Phone_ Number	Marks_ Maths	Marks_ Physics	Books_ Taken*	7																	
RollNo	Student_ Name	Address	Phone_ Number	Marks_ Maths	Marks_ Physics	Books_ Taken*																				
27	What is Normalization? Explain Normalization process for 1NF, 2NF and 3NF with example. (Dec 2014 Old) [SSIT]	7																								
28	What is Normalization? Normalize below given relation up to 3NF. (Nov 2017) [SSIT] STUDENT: <table border="1"><tr><td>Stu ID</td><td>Stu Name</td><td>City</td><td>Pin code</td><td>Project ID</td><td>Project Name</td><td>Course</td><td>Content</td></tr><tr><td>S101</td><td>Ajay</td><td>Surat</td><td>326201</td><td>P101</td><td>health</td><td>programming</td><td>C++, Java,C</td></tr><tr><td>S102</td><td>Vijay</td><td>Pune</td><td>325456</td><td>P102</td><td>social</td><td>WEB</td><td>HTML,PHP,ASP</td></tr></table>	Stu ID	Stu Name	City	Pin code	Project ID	Project Name	Course	Content	S101	Ajay	Surat	326201	P101	health	programming	C++, Java,C	S102	Vijay	Pune	325456	P102	social	WEB	HTML,PHP,ASP	7
Stu ID	Stu Name	City	Pin code	Project ID	Project Name	Course	Content																			
S101	Ajay	Surat	326201	P101	health	programming	C++, Java,C																			
S102	Vijay	Pune	325456	P102	social	WEB	HTML,PHP,ASP																			
29	What is meant by normalization? Write its need. List and discuss database anomaly during database design.(Dec 2018) [SSIT]	04																								
30	Normalize the following schema, with given constraints, to 4NF. <i>books</i> (<i>accessionno</i> , <i>isbn</i> , <i>title</i> , <i>author</i> , <i>publisher</i>) <i>users</i> (<i>userid</i> , <i>name</i> , <i>deptid</i> , <i>deptname</i>) List of constraints;	04																								

	<i>accessionno → isbn isbn → title isbn → publisher isbn →→ author userid → name userid → deptid deptid → deptname (june 2019)[SSIT]</i>	
31	What is Normalization? Explain 1NF, 2NF, 3NF, BCNF with example. (Nov-2019_Old) [SSIT]	7
32	Explain various Normal forms up to 3NF.(Nov-2019) [SSIT]	3
33	Explain Armstrong's Axioms in detail.(Nov-2019) [SSIT]	4
34	A software contract and consultancy firm maintain details of all the various projects in which its employees are currently involved. These details comprise: Employee Number, Employee Name, Date of Birth, Department Code, Department Name, Project Code, Project Description, Project Supervisor Assume the following: • Each employee number is unique. • Each department has a single department code. • Each project has a single code and supervisor. • Each employee may work on one or more projects. • Employee names need not necessarily be unique. • Project Code, Project Description and Project Supervisor are repeating fields. Normalize this data to Third Normal Form..(Nov-2019) [SSIT]	7
	TOPIC:3: Multi- valued dependency, 4NF, Join dependency and 5NF	
Sr. No	DESCRIPTIVE QUESTIONS	Marks
1	What are Multi-valued dependencies? Explain with an appropriate example (Dec 2014) [SSIT]	3
2	Why do we need normalization? Explain 4NF & 5NF. (May 2012) [SSIT]	7
	UNIT NO - 5 :Query Processing & Query Optimization	
	TOPIC:1 :Overview, measures of query cost, selection operation, sorting, join, evaluation of expressions, transformation of relational expressions estimating statistics of expression results, evaluation plans, materialized views	
	SHORT QUESTIONS	
1	What is called query processing ?) [SSIT]	1
2	Procedure of choosing a suitable query out of all queries is classified as _____. [a] query optimization [b] parser optimization [c] processing optimization [d] all of above ANS: d) [SSIT]	1
3	Representation of query in form of data structure is classified as _____. [a] query graph [b] query tree [c] scanner tree [d] parser tree ANS: b) [SSIT]	1
4	What are steps involved in query processing?) [SSIT]	1
5	Define query optimization.) [SSIT]	1
6	What is called an evaluation primitive?[SSIT]	1
7	What is called query execution engine? [SSIT]	1
8	How do you measure the cost of query evaluation? [SSIT]	1
9	What are called as index scans? [SSIT]	1
10	What are called as index scans? [SSIT]	1
11	What is called as external sorting? [SSIT]	1
12	Explain nested loop join? [SSIT]	1
13	What is called as recursive partitioning?[SSIT]	1

14	What is called as N-way merge?[SSIT]	1
15	Define Materialize view. (Nov-2019_Old) [SSIT] Answer: A materialized view is a database object that contains the results of a query. The FROM clause of the query can name tables, views, and other materialized views. Collectively these objects are called master tables (a replication term) or detail tables (a data warehousing term).	1
Sr. No	DESCRIPTIVE QUESTIONS	Marks
1	Explain query optimization process. Explain (Dec 2009 ,Dec 2014) [SSIT]	4
2	Explain evaluation of expression process in query optimization? (Dec 2009 ,Dec 2013) [SSIT]	4
3	Explain transformation of relational expression into equivalent relational expression. (Mar 2010) [SSIT]	7
4	Explain various steps involved in Query evaluation. (Mar 2010) [SSIT]	7
5	Explain linear search and binary search algorithm for selection operation.(Dec 2010) [SSIT]	7
6	Explain method of query optimization (May 2011,Jan 2017 Old) [SSIT]	4 , 6
7	Explain various steps of query evaluation.(May 2011) [SSIT]	4
8	Explain Query Optimization Process. (May 2012) [SSIT]	4
9	Explain the purpose of sorting with example with reference to query optimization (May 2012)(june 2019)[SSIT]	5
10	Explain the measures of finding out the cost of a query in query processing (May 2013)(june 2019)[SSIT]	5,4
11	Explain the measures of query cost, selection operation and join. (Jan 2013) [SSIT]	7
12	Write short note on query processing. (May 2014) [SSIT]	7
13	Enlist and explain the basic steps in Query Processing.(Dec 2015) [SSIT]	7
14	Assuming worst case memory availability and the following given statistics for 04 the relations customer and depositor Number of records of customer: 10,000 depositor: 5000 Number of blocks of customer: 400 depositor: 100 Estimate the cost i) with depositor as outer relation ii) with customer as the outer relation (Dec 2015) [SSIT]	4
15	Discuss various steps of query processing with proper diagram (May 2016) [SSIT]	7
16	Explain Query Processing steps.(Jun 2017) [SSIT]	4
17	Explain Heuristics in Optimization.(Jun 2017) [SSIT]	3
18	Explain steps of query processing with the help of neat diagram. (Jan 2017 ,May 2018) [SSIT]	7
19	Write a note on query processing with suitable diagram. (May 2016 Old) [SSIT]	7
20	What is query optimization? Explain how to optimize nested sub queries. (Dec 2014 Old) [SSIT]	7
21	Explain the steps in query processing? How the cost of query is measured? (Dec 2014 Old) [SSIT]	7
22	How expressions are evaluated? Or Explain materialization and pipelining.[SSIT]	4
23	Discuss various steps of query processing with proper diagram. (Nov 2017) [SSIT]	7
24	Explain query processing step with diagram.(Dec 2018) [SSIT]	7
25	Explain query evaluation process. (june 2019) [SSIT]	3
26	Discuss various steps of Query processing with diagram. (Nov-2019_Old,Nov-2019) [SSIT]	7,4

	UNIT NO - 6 : Storage strategies	
	TOPIC:1 Indices, B-trees, hashing	
1	Explain B-tree in details.[SSIT]	5
2	Explain insertion/deletion operation in b+ tree[SSIT]	7
3	Explain difference between b-tree and b+ tree[SSIT]	5
4	Write a Short note on hashing[SSIT]	05
5	Differentiate static hashing and dynamic hashing.[SSIT]	05
6	Construct a Btree for the following set of key values: (2,3,5,7,11,17,19,23,29,31) Assume that the tree is initially empty and values are added in ascending order. Consider the number of pointers in each node as four. (Nov 2019) [SSIT]	07
7	Explain Dense and Sparse indices in detail. (Nov 2019) [SSIT]	07
	UNIT NO - 7 : Transaction processing	
	TOPIC:1 :Transaction concepts, properties of transactions, serializability of transactions, testing for serializability	
Sr. No	SHORT QUESTIONS (1 Mark) / MCQ / True-False/Fill in the blanks	Marks
1	What is serialisability? what is its objective?(Jun 2017) [SSIT] Answer: Concurrent schedule produces result as of serial schedule	1
2	Define Transaction (Dec 2015, Nov-2019_Old)[SSIT] Answer: A transaction is a sequence of operations performed as a single logical unit of work.	1
3	What do you mean by concurrent execution of transaction? (May 2016)[SSIT] Answer: Where multiple transactions are being executed by the operating system in a multiprogramming environment, there are possibilities that instructions of one transaction are interleaved with some other transaction.	1
4	Give name of method to check the conflict serializability(May 2016)[SSIT] Answer: Precedence Graph method	1
5	Define Transaction. Dec 2015 .[SSIT]	1
6	Give name of method to check the conflict serializability. (May 2016) .[SSIT]	1
7	What is transaction? [SSIT]	1
8	What are the properties of transaction?[SSIT]	1
9	What are conflict serializable schedules? [SSIT]	1
10	What are conflict equivalent schedules?[SSIT]	1
11	What is view equivalence?[SSIT]	1
12	What is view serializable?[SSIT]	1
Sr. No	DESCRIPTIVE QUESTIONS	Marks
1	Explain ACID properties of transaction? (Dec 2009) [SSIT]	7
2	Explain working of 2-phase commit protocol. (Mar 2010) [SSIT]	4
3	Explain working of 2-phase commit protocol. (Mar 2010) [SSIT]	6
4	List the ACID properties. Explain the usefulness of each (Dec 2010) [SSIT]	5
5	Explain view serializability with example.. (Mar 2010) [SSIT]	5

7	Explain conflict serializability with example.. (Mar 2010) [SSIT] Write a note on conflict serializability. (May 2018, Dec 2018) [SSIT] Explain Conflict serializability. (Nov-2019_Old) [SSIT] Explain conflict serializability with the help of suitable example.(Nov 2019) [SSIT]	7 4 3
8	Explain Two-Phase commit protocol. (May 2011) [SSIT]	7
9	What are ACID properties? (Dec 2011) [SSIT]	4
10	Explain briefly the meaning of serializability of transactions. (May 2011) [SSIT]	7
11	Explain both the forms of serializability & relation between the two (May 2012) [SSIT]	2
12	What is a transaction? Explain the ACID properties. (Jan 2013) [SSIT]	7
13	Explain ACID properties of transaction. (Nov-2013) [SSIT] Explain ACID Properties of Transaction.(Nov-2019_Old) [SSIT] Enlist and explain ACID properties for transaction.(Nov 2019) [SSIT]	4 4 4
14	What is a transaction? Explain its four important properties. (May 2013, Jan 2017 Old) [SSIT] What is transaction? List and explain ACID property of transaction. (June 2019)[SSIT]	4, 6
15	What is a Transaction? Explain the properties of the transaction. Explain the States of the transaction with a neat sketch. (Dec 2014) [SSIT]	4
16	During its execution, a transaction passes through several states, until it finally commits or aborts. List all possible sequences of states through which a transaction may pass. Explain why each state transition may occur. (Dec 2014) [SSIT]	4
17	Explain conflict serializability and view serializability. (May 2014)(June 2019)[SSIT]	7
18	Explain ACID properties of transaction. (Dec 2015, May 2018) [SSIT]	4,4
19	Define Transaction . Explain the transaction properties and transaction states.(May 2015) [SSIT]	5
20	Define transaction. Explain various states of transaction with suitable diagram. (May 2016 , May 2018, Dec 2018)[SSIT]	3,4
21	Explain ACID properties of transaction. (May 2016) [SSIT]	4
22	Write a note on view serializability. (May 2016) [SSIT]	7
23	Explain Serializability with its types. (Jan 2017 Old)[SSIT]	6
24	Explain two phase locking protocol in detail.(Jun 2017) [SSIT]	7
25	Define Transaction. List out properties of transaction. Explain two-phase commit protocol.(Jun 2017) [SSIT]	4
26	State differences between conflict serializability and view serializability.(Jan 2017) [SSIT]	4
27	List and discuss ACID properties of transaction. (Jan 2017, Dec 2018) [SSIT]	4
28	Draw and explain state diagram of transaction. (May 2016 Old)[SSIT]	7
29	Define Serializability. Write a note on conflict serializability. (May 2016 Old)[SSIT]	7
30	Define Transaction. Discuss ACID properties of a transaction with suitable example.(May 2016 Old)[SSIT]	7
31	Explain ACID Properties of transaction with appropriate example. (Dec 2015 Old)[SSIT]	7
32	What is transaction? Draw and explain life cycle of transaction. (May 2015 Old)[SSIT]	7
33	What is serializability? Explain view serializability with example (Dec 2014 Old) [SSIT]	7
34	What is transaction?[SSIT]	3
35	Explain different states of transaction.[SSIT]	4
36	Explain the ACID properties using suitable example. (Nov 2017) [SSIT]	7
37	What do you mean by atomicity and aggregation?(Nov 2019_Old) [SSIT]	4
	TOPIC:2 :concurrent executions of transactions and related problems, Locking mechanism, solution to concurrency related problems, deadlock, , two-phase locking protocol, Isolation, Intent locking , Multi-version and optimistic Concurrency Control schemes	

Sr. No	SHORT QUESTIONS (1 Mark) / MCQ / True-False/Fill in the blanks	Marks
1	What is locking? Define each types of locking. (Jun 2017) [SSIT] Answer: To access shared object Lock is required. There are two types of lock Shared Mode: to access object in read only mode Exclusive mode: to access object in read/write mode	1
2	What is lock? [SSIT]	1
3	What are various types of locking techniques?[SSIT]	1
4	What is binary lock?[SSIT]	1
5	What is shared lock?[SSIT]	1
6	What is two phase locking?[SSIT]	1
1	What do you mean by concurrent execution of transaction? (May 2016) .[SSIT]	1
2	What is durability in DBMS? [SSIT]	1
3	What is recovery management component? [SSIT]	1
4	Give the reason for allowing concurrency? [SSIT]	1
5	Locking can not be used for _____. [a] deadlock [b] lost update [c] uncommitted dependency [d] inconsistent data ANS: a [SSIT]	1
6	_____ will undo all statements up to commit. [a] transaction [b] flashback [c] Rollback [d] Abort ANS: c [SSIT]	1
7	What is deadlock? [SSIT]	1
8	Define phases of two phase locking protocol.[SSIT]	1
9	What is Phantom Deadlock?[SSIT]	1
10	What is checkpoint and when it occurs?[SSIT]	1
11	What are two methods for dealing deadlock problem .[SSIT]	1
Sr. No	DESCRIPTIVE QUESTIONS	Marks
1	Why concurrency control is needed??(Dec 2009) [SSIT]	3
2	Explain Two phase locking? (Dec 2009 ,Dec 2011) [SSIT]	3
3	Consider schedule S with transaction T1 and T2. T1 transfer Rs. 150 from account A to C and T2 adds Rs. 50 into account A. Prepare concurrent schedule with two phase locking protocol. (Mar 2010) [SSIT]	5
4	What is concurrency? What are the three problems due to concurrency? How the problems can be avoided, explain for one of the three problems. (May 2011) [SSIT]	7
5	What is concurrency? If not controlled where it can lead to? What are the methods to control concurrency? (Nov 2013) [SSIT]	7
6	Explain the dirty read problem (Dec 2015) [SSIT]	3
7	Explain Locked based protocol (Jan 2013, Dec 2015) [SSIT]	7
8	What is deadlock? What are the ways to prevent deadlock? (Dec 2015 Old)[SSIT]	7
9	List and explain various issues while transactions are running concurrently in DBMS. (May 2015 Old)[SSIT]	7
10	Explain time stamp based protocols. (May 2014 Dec 2014) [SSIT]	7
11	What is deadlock? Explain Wait-For-Graph (May 2011) [SSIT]	2
12	Explain deadlock with example (Dec 2015) [SSIT]	3
13	Explain deadlock detection. (Dec 2011) [SSIT]	4
14	Explain various deadlock prevention methods. (Mar 2010) [SSIT]	5
15	What is deadlock? When it is occurs and how to avoid it? (Dec 2010) [SSIT]	7
16	What is deadlock? What are its prevention and avoidance methods? (Nov 2013) [SSIT]	7
17	What is deadlock? Explain deadlock detection and prevention policies (May 2014) [SSIT]	7
18	What is deadlock? Explain necessary conditions for deadlock and methods for handling it. (May	7

	2015) [SSIT]	
19	What is deadlock? Illustrate the same using the Wait-For-Graph (Dec 2014) [SSIT]	7
20	Write differences between shared lock and exclusive lock. (May 2016 ,May 2018, Dec 2018)[SSIT]	3
21	Write a note on two phase locking protocol. (May 2016,May 2018, Dec 2018,Nov 2019) [SSIT]	4,4
22	Explain Two-Phase Locking protocol. (May 2011 May 2012) [SSIT]	5
23	Explain Two phase locking (Dec 2014) [SSIT]	5
24	Define Locking. Explain two phase locking protocol. (Jan 2013) [SSIT]	7
25	Explain Two Phase Locking protocol. What are its advantages and disadvantages? (May 2013) [SSIT]	7
26	Explain Strict two phase locking with its advantage and disadvantages. (Dec 2010) [SSIT]	7
27	Explain two phase locking protocol in detail (Dec 2015) [SSIT]	7
28	What is locking? Explain Implicit and Explicit locking in oracle. (Jun 2017 Old)[SSIT]	7
29	Explain deadlock with suitable example. (Jan 2017) [SSIT]	3
30	Define Locking. Write a note on two phase locking protocol. (May 2016 Old)[SSIT]	7
31	What is need of lock in DBMS? Explain shared lock and exclusive lock with the help of example. (May 2015 Old)[SSIT]	7
32	Explain two phase locking protocol. Also list advantages of this protocol. (May 2015 Old)[SSIT]	7
33	Explain three type of actions to be taken for recovery from deadlock. (Dec 2014 Old) [SSIT]	7
34	Explain wait-die and wound-wait for deadlock prevention. (Dec 2014 Old) [SSIT]	7
35	Explain system recovery.[SSIT]	4
36	What is deadlock of transaction? Explain wound wait technique for prevention of deadlock.(May 2018, Dec 2018) [SSIT]	3
37	Explain Two phase locking protocol. (Nov 2017) [SSIT]	7
38	Explain various types of LOCKs used in Lock base protocol for concurrency control. (Nov 2017) [SSIT]	3
39	Explain deadlock with suitable example. (Nov 2017) [SSIT]	4
40	What is a recoverable schedule? Why is recoverability of schedules desirable? Are there any circumstances under which it would be desirable to allow non-recoverable schedules? Explain your answer (june 2019)[SSIT]	07
41	Define wait-Die and Wound-wait.(Nov-2019_Old) [SSIT]	3
	TOPIC:3 : System recovery, Two- Phase Commit protocol, Recovery and Atomicity, Log-based recovery	
Sr. No	DESCRIPTIVE QUESTIONS	Marks
1	Explain Two phase commit protocol? (Dec 2009) [SSIT] Write short note on 2PC Protocol.(Nov 2019_Old) [SSIT]	6 3
2	Explain immediate database modification log based recovery method. Also explain role of check point in log base. (Mar 2010) [SSIT]	07
3	Explain the Immediate Database modification Log Based Recovery method with an appropriate example. Explain role of check point in Log base. (Dec 2014) [SSIT]	7
4	Explain system recovery procedure with check point record concept. (May 2011)(june 2019) [SSIT]	7,4
5	Briefly describe immediate database modification for log based recovery.(May 2016) [SSIT]	7
6	Explain log based recovery and mention all its types (May 2012) [SSIT]	3
7	What is System recovery? Explain two phase commit protocol.(Jan 2013,Jan 2017 Old) [SSIT]	7
8	Explain system recovery procedure with check point record.(May 2013) [SSIT]	7
9	Explain log-based recovery and shadow paging technique. (May 2014) [SSIT]	7

10	Define Failure? Explain Log based Recovery. (May 2015) [SSIT] Explain Log based recovery method. (June 2019)[SSIT]	7,4
11	What is database Recovery? Explain Shadow paging in detail (Jun 2017 Old)[SSIT]	7
12	What is concurrency? Explain concurrency control techniques. (Jun 2017 Old, Jan 2017 Old) [SSIT]	7
13	Explain three concurrency problem. How does the strict two-phase locking protocol solve three problems of concurrency? explain with example.(Jun 2017) [SSIT]	7
14	Explain deferred database modification for log based recovery. 07(Dec 2014 Old) [SSIT]	7
15	Write a note on two phase commit protocol. (Jan 2017 ,May 2018, Dec 2018) [SSIT]	3
16	Define Failure. Write a note on log based recovery. (Jan 2017) [SSIT]	7
17	Explain working of two phase commit protocol. (Dec 2015 Old)[SSIT]	7
18	Why we need to do recovery in DBMS? Explain various method use for that. (May 2015 Old)[SSIT]	7
19	What is log based recovery? Explain Immediate database modification technique for database recovery.(May 2018, Dec 2018) [SSIT]	3
20	Describe log-based recovery in brief. (Nov 2019) [SSIT]	4
UNIT NO - 8 : Database Security		
TOPIC:1 : Authentication, Authorization and access control, DAC, MAC and RBAC models,		
SHORT QUESTIONS		
1	What are threats to database? .[SSIT]	1
2	What is the responsibility of database security and authorization subsystem? .[SSIT]	1
3	What is Access Control? .[SSIT]	1
4	List Different types of database security mechanism.[SSIT]	1
5	Explain purpose and scope of database security .[SSIT]	1
6	_____ is responsible for overall database security [a] DBA [b] DA [c] Programmer [D] Tester .[SSIT]	1
7	What is discretionary access control? [SSIT]	1
8	What is mandatory access control? [SSIT]	1
9	Differentiate between DAC and MAC. [SSIT]	1
Sr. No	DESCRIPTIVE QUESTIONS	Marks
1	Define Discretionary access control Dec 2010 [SSIT]	2
2	What do you mean by security? What the while designing secure database? (May 2012) [SSIT]	4
3	Explain the difference between Discretionary access control and mandatory access control. (May 2011) [SSIT]	3
4	Explain mandatory access control of database security? (Dec 2009) [SSIT]	3
5	Explain in detail Discretionary access control and mandatory access control (May 2013) [SSIT]	7
6	Explain the differences between Discretionary access control and mandatory access control (Dec 2014) [SSIT] Compare Discretionary access control and Mandatory Access Control.(Nov 2019_Old)[SSIT]	3 4

7	What is the difference between security and integrity? (May 2011) [SSIT]	2
8	What is authorization and authentication? Explain the access controls in a database. (Jan 2013) [SSIT] OR Explain Authorization and access control in brief. . (Nov 2019) [SSIT]	7 3
9	What is difference between authorization and authentication? Explain the use of audit trail. (Jun 2017) (May-2012) [SSIT]	3
10	Write & explain two rules in the Bell-LaPadula model for mandatory access control.(Jun 2017) [SSIT]	3
11	What is the difference between data security and data integrity(june 2019)[SSIT]	3
	TOPIC:2: Data Encryption, Intrusion detection, SQL injection.	
Sr. No	DESCRIPTIVE QUESTIONS	Marks
1	Describe Data Encryption. (Dec 2010) [SSIT]	2
2	Explain Data encryption in brief? (Dec 2009) [SSIT]	3
3	What is security of data? Explain data encryption. (May 2011) [SSIT]	4
4	Explain Data encryption (Dec 2011) [SSIT]	5
5	Write short note on data encryption. (Mar 2010) [SSIT]	5
6	What is security of data? Illustrate Data encryption with an appropriate example (Dec 2014) [SSIT]	3
7	What is cryptography? Explain the difference between symmetric & asymmetric key cryptography.. (Dec 2015) [SSIT]	6
	UNIT NO - 9 : SQL Concepts	
	TOPIC:1 : Basics of SQL, DDL,DML,DCL, structure – creation, alteration, defining constraints – Primary key, foreign key, unique, not null, check, IN operator Functions - aggregate functions, Built-in functions –numeric, date, string functions, set operations, sub-queries, correlated sub-queries, Use of group by, having, order by, join and its types, Exist, Any, All , view and its types	
Sr. No	SHORT QUESTIONS (1 Mark) / MCQ / True-False/Fill in the blanks	Marks
1	What will be the output of following SQL statement? select substr('123456789',instr('abcabcabc','b'), 4) from dual; A.6789 B. 2345 C. 56789 D. 456789 (Jun 2017) [SSIT] Answer: B	1
2	Define Constraint. List out constraints.(Jun 2017) [SSIT] Answer: Constraints are conditions that data must always satisfy. Constraints: Entity level	1

	constraint- primary key, referential integrity constraint- foreign key, Check constraint, Domain	
3	_____ can be used to retrieve data from multiple tables. A. Embedded SQL. B. Dynamic SQL. C. Joins. D. Views.(Dec 2015)[SSIT]	1
4	_____ is an attribute, or set of attributes, that uniquely identifies a tuple within a relation A. Foreign key B. Superkey C. Matching key D. None of above (Jun 2017) [SSIT] Answer: B	1
5	SQL belongs to the category of _____. A. 2GL. B. 3GL. C. 4GL. D. 5GL. (Dec 2015)[SSIT]	1
6	To modify the students table and to add a primary key on the student_id Column, Which statement must be used to accomplishes this task? Note: The table is currently empty. (Dec 2015)[SSIT] A. Alter table students add primary key student_id; B. Alter table students add constraint primary key (student_id); C. Alter table students add constraint stud_id_pk primary key student_id; D. Alter table students add constraint stud_id_pk primary key (student_id);	1
7	New fields can be added to the created table by using _____ command. (Dec 2015) [SSIT] A. ALTER. B. SELECT. C. CREATE. D. UPDATE.	1
8	For which action the TO_DATE function can be used? A. To convert any date literal to a date. B. To convert any numeric literal to a date. C. To convert any character literal to a date. D. To convert any date to a character literal (Dec 2015)[SSIT]	1
9	Which command is used to change the name of table? (May 2016)[SSIT] Answer: RENAME TABLE allows you to rename an existing table in any schema	1
10	Write functionality of DESC command in SQL. (May 2016)[SSIT] Answer: UsedTo sort list in descending order	1
11	Give full form of SQL and DCL (May 2016)[SSIT] Answer: SQL Structured Query Language DCL data control language	1
12	_____ SQL command is used to change or modify data values in the table. (Jan 2017) [SSIT] Answer: update	1
13	Which clause should be used to exclude group results? (Dec 2015)[SSIT] A. WHERE. B. HAVING. C. RESTRICT. D. GROUP BY.	1
14	The _____ function converts a value of DATE data type to CHAR date value. (Jan 2017) [SSIT] Answer: tochar()	1
15	State True or False: A table cannot contain multiple primary keys. (Jan 2017) [SSIT] Answer: TRUE	1
16	State True or False: SELECT DISTINCT statement is used to return only distinct values. (Jan 2017) [SSIT] TRUE	1

17	New column can be added to the existing table using _____ SQL command. (Jan 2017) [SSIT] Answer: Alter table with add column	1
18	_____ is a full form of SQL. (Jan 2017) [SSIT] Answer: Structured query language	1
19	The _____ operator is used to select values within a range. (Jan 2017) [SSIT] Answer: between	1
20	DDL stands for _____.(Jan 2017) [SSIT] Answer: Data description language	1
Sr. No	DESCRIPTIVE QUESTIONS	Marks
1	State the differences between COMMIT and ROLLBACK commands. May 2016 [SSIT]	3
2	Explain the Rollback and commit commands Dec 2014 [SSIT]	2
3	Explain commit and rollback command. Dec 2011 [SSIT]	5
4	Define DBMS. Explain the DBMS languages with examples: DDL, DML, and DCL May 2015 [SSIT]	7
5	Explain commands: (1) Rename (2) Alter (3) View (4) Unique (5) Drop (6) Default (7) Update (Jun 2017 Old)[SSIT]	7
6	Differentiate Between: (1) update and insert (2) primary key and foreign key (3) grant and revoke (4) row and column May 2015 [SSIT]	7
7	Explain following terms with suitable example. (1) Primary Key (2) Candidate Key (3) Foreign Key (4) Check Constraint Dec 2015 [SSIT]	4
8	What is NULL? Explain May 2012 [SSIT]	2
9	Describe GRANT and REVOKE commands. (May 2016,May 2018, Dec 2018) [SSIT]	3
10	What are transaction control commands? Explain any two commands (May 2011) [SSIT]	2
11	Explain Transaction Control Commands. (May 2012) [SSIT]	3
12	Discuss aggregate functions with example(s). (May 2016) [SSIT]	3
13	Define view. What are the types of view? Write syntax to create view of each type. Give an example of view.(Jun 2017)[SSIT] Define view. Write syntax to create view .Give an example of view. (May 2018) [SSIT]	3
14	Explain any two aggregate functions of SQL. May 2013 [SSIT]	2
15	Explain any three Aggregate functions and Scalar functions with examples. (May 2015) [SSIT]	7
16	Explain Aggregate, Date and string functions with example. May 2014 [SSIT]	7
17	Explain DDL statements Dec 2014 [SSIT]	2
18	Explain DDL,DML,DCL with example Dec 2010 Dec 2011 May 2014 [SSIT]	5
19	Explain any two string functions in SQL. May 2013 [SSIT]	2
20	Write with example various built in string functions. (Dec 2010) [SSIT]	5
21	Explain any two string functions in SQL. (May 2011) [SSIT]	2
22	What is ON DELETE CASCADE in SQL? Explain clearly with example (May 2013) [SSIT] Explain on delete cascade with example. (june 2019)[SSIT]	4
23	What is View? Give the advantages of View. (May 2012) [SSIT]	2
24	What is a view? Explain how to create, its types and significance in DBMS. (May 2015) [SSIT]	7
25	What is a view? What are its types? Write the syntax for creating a view (May 2013) [SSIT]	7
26	What is a view? Explain the syntax. Explain the different types of views.(Dec 2014)[SSIT]	7

27	Write query for the following: (1) To create a table from a table. (2) To eliminate duplicate rows. (3) To add a new column in the table 4. To sort data in a table(May 2015) [SSIT]	7
28	Write query for the following. (1) Create a table from at table. (2) To join two existing tables. (3) To view system date. (4) To drop a view. (Jun 2017 Old)[SSIT]	7
29	What is materialized view? Explain Query optimization with materialized view.(Jun 2017)[SSIT]	4
30	we have following relations: Supplier(S#,sname,status,city) Parts(P#,pname,color,weight,city) SP(S#,P#,quantity) Answer the following queries in SQL. (i) Find name of supplier for city = 'Delhi'. (ii) Find suppliers whose name start with 'AB' (iii) Find all suppliers whose status is 10, 20 or 30. (iv) Find total number of city of all suppliers. (v) Find s# of supplier who supplies 'red' part. (vi) Count number of supplier who supplies 'red' part. (vii) Sort the supplier table by sname.?(Dec 2009) [SSIT]	7
31	Enlist SQL commands for DML. Consider following relations and write SQL queries for given statements. Assume suitable constrains. job(job-id, job-title, minimum-salary, maximum-salary) employee(emp-no, emp-name, emp-salary,dept-no) deposit(acc-no, cust-name, branch-name, amount, account-date) borrow(loan-no, cust-name, branch-name, amount) department (dept-no, dept-name) 1. Give name of depositors whose branch name starts from 'D'. 2. Give employee name(s) whose salary is between Rs. 20000 to 30000 and department name is Finance 3. Delete borrower details whose amount is less than 25000. (May 2016) [SSIT]	7
32	We have following relations: EMP(empno, ename, jobtitle, managerno, hiredate, sal, comm, deptno) DEPT(deptno, dname, loc) Answer the following queries in SQL. i) Find the Employees working in the department 10, 20, 30 only. ii) Find Employees whose names start with letter A or letter a. iii) Find Employees along with their department name. iv) Insert data in EMP table. v) Find the Employees who are working in Smith's department vi) Update Department name of Department No=10 vii) Display employees who are getting maximum salary in each department (Dec 2015)[SSIT]	7
33	we have following relations: Supplier(S#,sname,status,city) Parts(P#,pname,color,weight,city) SP(S#,P#,quantity) Answer the following queries in SQL.	7

	(i) Delete records in supplier table whose status is 40. (ii) Add one field in supplier table. (iii) Explain commit command (iv) Explain Curser in PL/SQL.? (Dec 2009) [SSIT]	
34	we have following relations: Supplier(S#,sname,status,city) Parts(P#,pname,color,weight,city) SP(S#,P#,quantity) Answer the following queries in SQL. (i) Find name of parts whose color is 'red' (ii) Find parts name whose weight less than 10 kg. (iii) Find all parts whose weight from 10 to 20 kg. (iv)Find average weight of all parts. (v) Find S# of supplier who supply part 'p2' (vi) Find name of supplier who supply maximum parts. (vii) Sort the parts table by pname.? (Dec 2009) [SSIT]	7
35	Supplier(S#,sname,status,city) Parts(P#,pname,color,weight,city) SP(S#,P#,quantity) Answer the following queries in SQL. (i) Find name of parts whose color is 'red' (ii) Find parts name whose weight less than 10 kg. (iii) Find all parts whose weight from 10 to 20 kg. (iv)Find average weight of all parts. (v) Find S# of supplier who supply part 'p2' (vi) Find name of supplier who supply maximum parts. (vii) Sort the parts table by pname.? (Dec 2009) [SSIT]	7
36	we have following relations: Supplier(S#,sname,status,city) Parts(P#,pname,color,weight,city) SP(S#,P#,quantity) Answer the following queries in SQL. (i) Delete records in parts table whose color is 'blue'. (ii) Drop one field in parts table. (iii) Explain rollback command. (iv) Explain stored procedure in PL/SQL.?(Dec 2009) [SSIT]	7
37	we have following relations: Supplier(S#,sname,status,city) Parts(P#,pname,color,weight,city) SP(S#,P#,quantity) Answer the following queries in SQL. (i) Delete records in parts table whose color is 'blue'. (ii) Drop one field in parts table. (iii) Explain rollback command. (iv) Explain stored procedure in PL/SQL.? (Dec 2009) [SSIT]	8
38	Write query for the following. (1) Create a table from at table. (2) To join two existing tables. (3) To view system date. (4) To drop a view. (Jun 2017 Old)[SSIT]	7
39	List the aggregate function available in SQL .Explain each aggregate function with example in brief. (Dec 2015 Old)[SSIT]	7
40	Write down the query for the following table where primary keys are underlined. Person(ss#, name, address) Car(license, year, model) Accident(date, driver, damage-amount) Owns(ss#, license)	8

	Log(license, date, driver) (1) Find the total number of people whose cars were involved in accidents in 2009. (2) Find the number of accidents in which the cars belonging to “S.Sudarshan” (3) Add a new customer to the database. (4) Add a new accident recorded for the Santro belonging to “KORTH” (Dec 2010)[SSIT]	
41	Consider the employee data. Give an expression in SQL for the following query : Employee(employee-name, street,city) Works(employee-name, company-name,salary) Company(company-name, city) Manages(employee-name, manager-name) (1) Find the name of all employees who work for State Bank. (2) Find the names and cities of residence of all employees who work for State Bank. (3) Find all employees in the database who do not work for State Bank. (4) Find all employees in the database who earn more than every employee of UCO Bank.(Dec 2010) [SSIT]	8
42	Consider following schema and write SQL for given statements. Student(Rollno, Name, Age,Sex,City) Student_marks(Rollno,Sub1,Sub2,Sub3,Total,Average) Write query to (i) Calculate and store total and average marks from Sub1, Sub2& Sub3. (ii) Display name of students who got more than 60 marks insubject Sub1. (iii) Display name of students with their total and average marks. (iv) display name of students who got equal marks in subject Sub2 (May 2011) [SSIT]	8
43	(a) Implement following relation using SQL query. Student(stud_no,stud_name,sub1,sub2,totalmark,percentage) Create the table, add 5 records and display the data (b) Calculate total mark and percentage and also arrange the students on ascending order of total mark and also make a view of it. (c) Update the mark of sub1 of student_no=111 with 50 and also Calculate totalmark and percentage accordingly. (Dec 2011) [SSIT]	5
44	Implement following relation using SQL query. Employee(emp_no,emp_name,department,city,salary) (1) Find all the employee whose emp_no is less than 100 and salary more than 25000 and department is “Account” (2) count the no of employee and Sum the salary of all employee (3) Delete the employee having minimum salary. (Dec 2011) [SSIT]	5
45	i) The employees who are getting salary greater than 3000 for those persons belonging to the department 20 ii) Employees who are not getting any commission iii) Find how many job titles are available in employee table. iv) Display total salary spent for each job category. v) Display number of employees working in each department and their department name. vi) List ename whose manager is NULL. vii)List all employee names and their salaries, whose salary lies between 1500/- and 3500/- both	7

	inclusive. (May 2012) [SSIT]	
46	we have following relations EMP(empno, ename, jobtitle, managerno, hiredate, sal, comm, deptno) DEPT(deptno, dname, loc) Answer the following queries in SQL. i) Find the Employees working in the department 10, 20, 30 only. ii) Find Employees whose names start with letter A or letter a. iii) Find Employees along with their department name. iv) Find Employees whose manager is KING. v) Find the Employees who are working in Smith's department vi) Find the Employees who get salary more than Allen's salary. vii) Display employees who are getting maximum salary in each department (june 2019)(May 2012) [SSIT]	7
47	Write queries for the following tables. T1 (Empno, Ename , Salary, Designation), T2 (Empno, Deptno.) (1) Display all rows for salary greater than 5000. (2) Display the deptno for the ename='syham'. (3) Add a new column deptname in table T2. (4) Change the designation of ename='ram' from 'clerk' to 'senior clerk'. (5) find the total salary of all the rows. (6) Display Empno, Ename, Deptno and Deptname. (7) Drop the table T1. (Jan 2013) [SSIT]	7
48	Consider following schema and write SQL for given statements. Student(RollNo, Name, Age, Sex, City) Student_marks(RollNo, Sub1, Sub2, Sub3, Total, Average) Write query to (i) Display name and city of students whose total marks are greater than 225. (ii) Display name of students who got more than 60 marks in each subject. (iii) Display name of city from where more than 10 students come from. (iv) Display a unique pair of male and female students. (May 2013) [SSIT]	8
49	Solve following queries with following table, where underlined attribute is primary key. Person(ss#, name, address) Car(license, year, model) Accident(date, driver, damage-amount) Owns(ss#, license) Log(license, date, driver) 1. Find the name of a person whose license number is '12345'. 2. Display name of driver with number of accidents done by that driver. 3. Add a new accident by 'Ravi' for 'BMW' car on 01/01/2013 for damage amount of 1.5 lakh rupees. (Nov 2013) [SSIT]	8
50	For Supplier – Parts database Supplier(S#,sname,status,city) Parts(P#,pname,color,weight,city) SP(S#,P#,quantity) Answer the following queries in SQL. 1. Display the name of supplier who lives in 'Ahmedabad'.	8

	2. Display the part name which is not supplied yet. 3. Count how many times supplier 'S1' has supplied part 'P1'. 4. Find all suppliers whose status is either 20 or 30. (Nov 2013) [SSIT]	
51	For Supplier – Parts database Supplier(S#,sname,status,city) Parts(P#,pname,color,weight,city) SP(S#,P#,quantity) Answer the following queries in SQL. 1. Find the name of part having 'Red' colour. 2. Delete parts whose weight is more than 100 gram. 3. Count how many times each supplier has supplied part 'P2'. 4. How much time shipment is for more than 100 quantities? (Nov 2013) [SSIT]	8
52	Write queries for the following tables: T1 (Empno, Ename , Salary, Designation) T2 (Empno, Deptno.) (1) Display all the details of the employee whose salary is lesser than 10K. (2) Display the Deptno in which Employee Seeta is working. (3) Add a new column Deptname in table T2. (4) Change the designation of Geeta from 'Manager' to 'Senior Manager'. (5) Find the total salary of all the employees. (6) Display Empno, Ename, Deptno and Deptname. 7) Drop the table T1. (Dec 2014) [SSIT]	7
53	Write SQL Queries for following set of tables: EMPLOYEE (EmpNo, Name, DoB, Address, Gender, Salary, DNumber) DEPARTMENT (DNumber, Dname, ManagerEmpNo, MnagerStartDate). 1. Display the Age of 'male' employees. 2. Display all employees in Department named 'Marketing'. 3. Display the name of highest salary paid 'female' employee. 4. Which employee is oldest manger in company? 5. Display the name of department of the employee 'SMITH'. (Jan 2017 Old)[SSIT]	7
54	Explain with example: (1) order by (2) group by and having (3) as select (4) schema (Jun 2017 Old)[SSIT]	8
55	Explain with reference to SQL: 1. Group by & having 2. Commit, rollback, savepoint. (Jan 2017 Old)[SSIT]	8
56	Consider the following employee database, primary keys are underlined : employee (employee-name, <u>street</u>, city) works (employee-name, company-name, salary) company (company-name, <u>city</u>) manages (employee-name, <u>manager_name</u>) Give an expression in SQL for each of the following queries: 1. Find the names of all employees who work for First Bank Corporation. 2. Give all employees of First Bank Corporation a 10-percent raise. 3. Find the names and cities of residence of all employees who work forFirst Bank Corporation.	7

	4. Find the names, street addresses, and cities of residence of all employees who work for First Bank Corporation and earn more than \$10,000. 5. Find all employees in the database who live in the same cities as the companies for which they work. 6. Find all employees in the database who do not work for First Bank Corporation. 7. Find the company and number of employees in company that has more than 30 employees. (Jun 2017) [SSIT]	
57	Consider the following Hotel database, primary keys are underlined : hotel (hotel-no, name, address) room (room-no, hotel-no, type, price) booking (hotel-no, guest-no, date-from, date-to, room-no) guest (guest-no, name, address) Give an expression in SQL for each of the following queries: 1. List the names and addresses of all guests in London, alphabetically ordered by name. 2. List out hotel name and total number of rooms available. 3. List the details of all rooms at the Grosvenor Hotel, including the name of the guest staying in the room, if the room is occupied. 4. List all guests currently staying at the Grosvenor Hotel. 5. List the rooms that are currently unoccupied at the Grosvenor Hotel. 6. List the number of rooms in each hotel in London. 7. List out all guests who have booked room for three or more days. (Jun 2017) [SSIT]	7
58	Explain following SQL*Plus commands with suitable example 1. Commit 2. Grant 3. Rollback 4. Revoke (Nov 2013) [SSIT]	7
60	Write SQL Queries for following set of tables: EMPLOYEE (EmpNo, Name, DoB, Address, Gender, Salary, DNumber) DEPARTMENT (DNumber, Dname, ManagerEmpNo, MmanagerStartDate). 1. Display the Age of 'male' employees. 2. Display all employees in Department named 'Marketing'. 3. Display the name of highest salary paid 'female' employee. 4. Which employee is oldest manager in company? 5. Display the name of department of the employee 'SMITH'. (Jan 2017 Old) [SSIT]	7
61	Explain with example: (1) order by (2) group by and having (3) as select (4) schema (Jun 2017 Old) [SSIT]	7
62	Explain with reference to SQL: 1. Group by & having 2. Commit, rollback, savepoint. (Jan 2017 Old) [SSIT]	8
63	Consider the following employee database, primary keys are underlined : employee (employee-name, street, city) works (employee-name, company-name, salary) company (company-name, city) manages (employee-name, manager_name) Give an expression in SQL for each of the following queries: 1. Find the names of all employees who work for First Bank Corporation.	7

	2. Give all employees of First Bank Corporation a 10-percent raise. 3. Find the names and cities of residence of all employees who work for First Bank Corporation. 4. Find the names, street addresses, and cities of residence of all employees who work for First Bank Corporation and earn more than \$10,000. 5. Find all employees in the database who live in the same cities as the companies for which they work. 6. Find all employees in the database who do not work for First Bank Corporation. 7. Find the company and number of employees in company that has more than 30 employees. (Jun 2017) [SSIT]	
64	Consider the following Hotel database, primary keys are underlined : hotel (hotel-no, name, address) room (room-no, hotel-no, type, price) booking (hotel-no, guest-no, date-from, date-to, room-no) guest (guest-no, name, address) Give an expression in SQL for each of the following queries: 1. List the names and addresses of all guests in London, alphabetically ordered by name. 2. List out hotel name and total number of rooms available. 3. List the details of all rooms at the Grosvenor Hotel, including the name of the guest staying in the room, if the room is occupied. 4. List all guests currently staying at the Grosvenor Hotel. 5. List the rooms that are currently unoccupied at the Grosvenor Hotel. 6. List the number of rooms in each hotel in London. 7. List out all guests who have booked room for three or more days. (Jun 2017) [SSIT]	7
65	Define: Primary key, Foreign Key and NOT NULL constraint. (Jan 2017 ,May 2018) [SSIT]	3
66	List and explain DML statements with suitable example. (Jan 2017 ,May 2018) [SSIT]	3
67	Write SQL statements (Query) for following tables: T1(rollno, stuname, age, city, branchcode) T2(branchcode, branchname) 1. Retrieve students details whose branchcode is 5. 2. Find an average age of all students. 3. Add new branch in T2 table. 4. Display rollno, stuname and age of students whose city is Chennai. 5. Change age of student to 20 whose rollno is 1. 6. Delete student details whose age is 18. 7. Retrieve branch information in descending order. (Jan 2017) [SSIT]	7
68	List and explain aggregation functions with suitable example. (Jan 2017 ,May 2018, Dec 2018) [SSIT]	4
69	Write differences between grant and revoke commands with suitable example. (Jan 2017) [SSIT] Differentiate Grant and Revoke with example. (Nov-2019_Old) [SSIT]	4
70	Compare rollback with commit SQL commands. (Jan 2017 ,May 2018) [SSIT]	4
71	Consider following given relations. Assume suitable primary keys for given relations. job(job-id, job-title, minimum-salary, maximum-salary) employee(emp-no, emp-name, emp-salary, dept-no) deposit(acc-no, cust-name, branch-name, amount, account-date) borrow(loan-no, cust-name, branch-name, amount) Write SQL queries for following given statements.	7

	1. Retrieve name of employee whose salary is between 10000 to 25000. 2. List total deposit amount of customer who belongs to 'Bangalore' branch. 3. Retrieve details of employee whose name begins with 'R'. 4. Modify employee name whose employee no. is 101. 5. Display distinct customer name whose loan number is greater than 5. 6. Insert new record of employee. 7. Change the name of borrow table to new-borrow table.(May 2016 Old)[SSIT]	
72	Write a note on DML and DDL statements with appropriate example.(May 2016 Old)[SSIT]	7
73	Explain commit and rollback commands with suitable example.(May 2016 Old)[SSIT]	7
74	we have following relations: Vender(V#,vname,status,city) [Note : Status can have two values : Active,Passive], Parts(P#,pname,code,weight,city),VP(V#,P#,quantity) Answer the following queries in SQL. A. Find name of Vender for city = 'Ahmedabad'. B. Find Venders whose name start with 'D' C. Find all Venders whose status is "Active". D. Find total number of city of all Venders. E. Find V# of Venders who supplies 'code = AXR123' part. F. Count number of Venders who supplies 'red' parts. G. Sort the parts table by pname in ascending order. (Dec 2015 Old)[SSIT]	7
75	What is aggregate function? Explain all aggregate functions with the help of example. (May 2015 Old)[SSIT]	7
76	Explain Having, Group by and order by clause with example. (Dec 2014 Old) [SSIT]	7
77	What is aggregate function? Explain any five aggregate functions with example. (Dec 2014 Old) [SSIT]	7
78	Explain Inner Join and Outer Join with SQL queries. (Dec 2014 Old) [SSIT]	7
79	Explain various operations in DDL, DML and DCL SQL languages. 07(Dec 2014 Old) [SSIT]	7
80	Explain SELECT & CREATE statement in SQL.[SSIT]	4
81	Explain DROP, UPDATE & DELETE statement in SQL.[SSIT]	5
82	What is ON DELETE CASCADE in SQL? Explain clearly with example[SSIT]	4
83	Consider following schema and write SQL for given statements. Client_master(clientno, name, address, city, pincode, state, baldue)Product_master(productno, name, profitpercent, unitmeasure, sellprice, costprice)Salesman_master (Salesmanno,name,address,city,pincode, state,salary,tgtotget, remarks) 1) Find out the names of all clients. 2) List all the clients who are located in Mumbai. 3) Delete all salesmen from salesman_master whose salaries are equal to Rs.3500. 4) Destroy the table client_master along with data. 5) List the name of all clients having 'a' as the second letter in their names. 6) Count the number of products having cost price is less than or equal to 500. 7) Calculate the average, minimum and maximum sell price of product.(May 2018) [SSIT]	07
84	Describe GRANT and REVOKE commands with suitable example. (Nov 2017) [SSIT]	4
85	Explain Aggregate Functions of SQL with suitable example. (Nov 2017) [SSIT]	7
86	Define View. Explain with appropriate example. (Nov 2017) [SSIT]	4

87	Consider following relations and write SQL queries for given statements. Assume suitable constraints. Instructor(ID, Name, Dept_name , Salary) Teaches(ID, Course_id, Sec_id, Semester(even/odd),Year) 1) Write SQL query to create Instructor table. 2) Find the average salary of the instructors in computer department. 3) Find the number of instructors in each department who teach a course in evensemester of 2016. 4) Find the names of instructor with salary amounts between 30000 and 50000.(Nov 2017) [SSIT]	7
88	Write a PL/SQL program for inserting even numbers in EVEN table and odd number in ODD table from number 1 to 50. (Nov 2017) [SSIT]	7
89	Define primary key, NOT NULL key and Unique key.(Dec 2018) [SSIT]	03
90	Define DDL, DML and DCL.(Dec 2018) [SSIT]	03
91	List the reason why NULL values introduces might be introduces in database.(Dec 2018) [SSIT]	04
92	Consider following schema and write SQL for given statements. employee (employee-name, street, city) works (employee-name, company-name, salary) company (company-name, city) manages (employee-name, manager_name) 1. Find the names of all employees who work for First Bank Corporation. 2. Give all employees of First Bank Corporation a 10-percent raise. 3. Find the names and cities of residence of all employees who work for First Bank Corporation. 4. Find the names, street addresses, and cities of residence of all employees who work for First Bank Corporation and earn more than \$10,000. 5. Find all employees in the database who live in the same cities as the companies for which they work. 6. Find all employees in the database who do not work for First Bank Corporation.(Dec 2018)[SSIT]	07
93	Describe the following SQL functions: (i) Add_months(x,y) (ii) Rtrim() (iii) To_date (june 2019)[SSIT]	03
94	Explain commit, rollback and savepoint command (june 2019)[SSIT] Define commit, rollback and save point.(Nov-2019_Old) [SSIT]	03
95	Explain Aggregate functions with example. (Nov-2019_Old)[SSIT]	7
96	Describe sub query and Correlated sub query.(Nov-2019_Old)[SSIT]	3
97	Define Join. Explain all Joins with examples.(Nov-2019_Old)[SSIT]	7
98	Define Encryption, DCL and DML.(Nov-2019_Old) [SSIT]	3
99	Write SQL statements (Query) for following tables: E1 (eno,ename,age,city,bcode) 2 (bcode, bname) 1. Display employees' details whose branch code is between 1 to 5. 2. Find an average age of all employees whose branch code is 3. 3. Retrieve branch information in descending order. 4. Add new branch in E2 table. 5. Display eno, ename and age of employees whose city is Ahmedabad. 6. Delete employee details whose age is 29. 7. Change age of employee to 25 whose employee no is 6.(Nov-2019_Old) [SSIT]	7
100	Consider following schema and write SQL for given statements.	7

	Student (RollNo, Name, DeptCode, City) Department (DeptCode, DeptName) Result (RollNo, Semester, SPI) 1. Display the name of students with RollNo whose name ends with 'sh'. 2. Display department wise total students whose total students are greater than 500. 3. List out the RollNo, Name along with CPI of Student. 4. Create RollNo field as primary key for existing Student table. 5. Display student name who got highest SPI in semester 1. 6. Display the list of students whose DeptCode is 5, 6,7,10. 7. Create table Student_New from student table without data. (Nov-2019) [SSIT]	
101	Consider the tables given below. Write the SQL queries for the questions given below: T1 (Empno, Ename , Salary, Designation,) T2 (Empno, Deptno.) (1) Display all the details of the employee whose salary is lesser than 10000. (2) Display the Deptno in which Employees with name starting with letter 'S' is working. (3) Add a new column Deptname in table T2. (4) Change the designation of Geeta from 'Manager' to 'Senior Manager'. (5) Find the total salary of all the employees department wise. (6) Add Empno as primary key in existing table T1. (7) Display the Deptno having highest number of employees. (Nov-2019) [SSIT]	7
	UNIT NO - 10 :PL/SQL Concepts :	
	TOPIC:1 :Cursors	
Sr. No	DESCRIPTIVE QUESTIONS	Marks
1	Explain the advantages of PL/SQL (May 2012) [SSIT]	3
2	Write a PL/SQL block to print the sum of even numbers from 1 to 100. (Dec 2014) [SSIT]	7
3	Write a PL/SQL block to print the sum of odd numbers from 1 to 100. (Dec 2014) [SSIT]	7
4	Define PL/SQL. Write PL/SQL block to print sum of even numbers between 1 to 20 (May 2016) [SSIT]	7
5	Write A PL/SQL block to print the sum of Numbers from 1 to 50. (May 2014) [SSIT] Write a PL/SQL block to print the sum of Numbers from 1 to 100. (june 2019)[SSIT]	7,3
6	Write A PL/SQL block to print the given number is Odd or Even (May 2014) [SSIT]	7
7	What is cursor? Explain with example. (Dec 2011) [SSIT]	5
8	Explain Cursor & it's type in PL/SQL. (Jan 2017 Old)[SSIT]	7
9	Define cursor. Briefly describe types of cursor in PL/SQL. (May 2016) [SSIT]	4
10	Write short note on cursors in PL/SQL. (May 2011) [SSIT]	5
11	Explain Cursor in PL/SQL (May 2012) [SSIT]	4
12	Write a PL/SQL cursor to display the names and branch of all students from the STUDENT relation. (Dec 2015) [SSIT]	4
13	What is cursor? Explain the types of cursors. (May 2013) [SSIT]	7

14	Write Note on Cursor and its types (Dec 2010 ,Dec 2013) [SSIT]	7
15	What is a cursor? Explain the built-in cursor types. (Jan 2013) [SSIT]	7
16	Write a PL/SQL block using explicit cursor that will display the customer name, the fixed deposit number and the fixed deposit amount of the first 5 customers holding the highest amount in fixed deposits. Use following database: cust_mstr (custno, name, occupation) fd_dtls (fd_ser_no, fd_no, type, period, opndt, duedt, amt, dueamt) acct_fd_cust_dtls (acct_fd_no, custno) (Jun 2017)[SSIT]	7
17	Discuss various types of cursors in PL/SQL.(Jan 2017) [SSIT]	3
18	Write PL/SQL code to print sum of even number between 1 to 100 numbers. (Jan 2017) [SSIT]	7
19	What is cursor? List and explain various types of cursors used in PL/SQL.(May 2016 Old)[SSIT]	7
20	Explain cursor in PL/SQL with example. (Dec 2015 Old)[SSIT]	7
21	Explain following with example(Any 4) PL/SQL block, Cursor, Trigger, Stored procedure, View(May 2015 Old)[SSIT]	14
22	What is non-procedural Language? Write a PL/SQL block to print the square of number from 1 to 10 (Inclusive). (Dec 2014 Old) [SSIT]	7
23	What is difference between SQL and PL/SQL? Write a PL/SQL block to determine the given number is odd or even. (Dec 2014 Old) [SSIT]	7
24	Consider wholesaler of book schema. Book (Book_id, book_title,publisher,book_price) (i) Implement procedure which print details of books whose price is more than average price.(Use cursor). (ii) Write a trigger such that if record is deleted from book table, inserts old record inbook_backup table. Mar 2010 [SSIT]	8
25	Explain Cursor in PL/SQL (Nov 2017) [SSIT]	3
26	Write a PL/SQL block to print the given number is prime or not. (june 2019)[SSIT]	3
27	Write PL/SQL block to print whether the given number is Armstrong number or not.(Nov 2019_Old) [SSIT]	7
28	Write short note on Cursor.(Nov 2019_Old) [SSIT]	3
29	What is PL/SQL. Explain the difference between SQL and PL/SQL.(Nov 2019) [SSIT]	3
	TOPIC:2 : Stored Procedures, Stored Function	
	SHORT QUESTIONS	
1	What is view in SQL? How is it defined? [SSIT]	1
2	List the table modification commands in SQL. [SSIT]	1

3	Database schema is written in _____. [a] HLL [b] DML [c] DDL [d] DCL ANS: c [SSIT]	1
4	_____ is a virtual table that draws its data from the result of an SQL SELECT statement. [a] View [b] Synonym [c] Sequence [d] Transaction ANS: a [SSIT]	1
5	ON UPDATE CASCADE ensures which of the following? [a] Normalization [b] Data Integrity [c] Materialized View [d] All of the above ANS: b [SSIT]	1
Sr. No	DESCRIPTIVE QUESTIONS	Marks
1	Explain stored procedures and stored functions. (Jan 2013) [SSIT]	7
2	Write a note on stored procedure with suitable example (May 2016) [SSIT] Write short note on stored Procedure with example.(Nov-2019_Old) [SSIT]	7 4
3	Explain Stored Procedure in PL/SQL(May 2012) [SSIT]	4
4	A stored function is created to perform the acct_no check operation. f_ChkAcctNo() is the name of function which accepts a variable acct_no from the user and returns value 0 if acct_no does not exist or 1 if acct_no exists. Write a PL/SQL block that performs transaction(i.e., deposit/withdrawal) on account. If account exists, change balance depending on the transaction amount to be deposited or withdrawal. Assume account table with fields – account number, name, type and balance. (Jun 2017) [SSIT]	3
5	Explain stored procedure with proper example. (Jan 2017 ,May 2018, Dec 2018) [SSIT]	7
6	Define PL/SQL. Write a note on stored procedure. (May 2016 Old)[SSIT]	7
7	What is Stored Procedure in PL/SQL? Write a Stored procedure to Insert Data in Student_Detail [Student_id,Name,Age,Class] table. (Dec 2015 Old)[SSIT]	7
TOPIC:3 :Database Triggers		
DESCRIPTIVE QUESTIONS		
1	Explain Database Triggers. (May 2012) [SSIT] Write a note on trigger (May 2018, Dec 2018) [SSIT]	3
2	What is trigger? Explain its type with their syntax. What are the applications of trigger? (Nov 2013) [SSIT]	7
3	Explain Trigger and its types. Explain its applications and also the syntax to create it (May 2013) [SSIT]	7
4	What are triggers? Explain the advantages and the needs (Dec 2014) [SSIT]	7
5	Write short note on database triggers in PL/SQL. (May 2011) [SSIT]	5
6	What is trigger? How it is useful? Explain (Dec 2011) [SSIT]	4
7	Explain Trigger with example (Dec 2010) [SSIT]	7

8	What is the Database Trigger? Explain it's use in DBMS.(Jan 2017 Old)[SSIT]	7
9	Write a note on trigger.(Jan 2017) [SSIT]	3
10	What are the types of trigger available in SQL? Write trigger to add Primary Key: Student_ID in Student_Master table. Student_ID is Number. (Dec 2015 Old)[SSIT]	7
11	What are triggers? Explain the advantages and the needs of triggers. (Nov 2017) [SSIT]	4